PHARVARIS

Study Title:	A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Orally Administered Deucrictibant Extended-release Tablet for Prophylaxis Against Angioedema Attacks in Adolescents and Adults with Hereditary Angioedema
Acronym:	CHAPTER-3
Study Number:	PHA022121-C305
Study Phase:	3
Product Name:	Deucrictibant 40 mg Extended-release Tablet
Indication:	Hereditary Angioedema
EU Trial Number:	2024-516247-62
IND Number:	153097
Investigators:	Multicenter
Sponsor:	Pharvaris Netherlands B.V. J.H. Oortweg 21 2333 CH Leiden The Netherlands
Sponsor Contact:	clinical@pharvaris.com
Sponsor Representative(s):	[redacted] Chief Medical Officer Pharvaris Netherlands B.V. [redacted] Clinical Development Lead, Executive Director Pharvaris Netherlands B.V.

Protocol Version	Date
Original Protocol	08 July 2024
Amendment 1.1 (US-specific)	26 August 2024
Amendment 1.2 (US-specific)	18 December 2024
Amendment 1.3 (EU-specific)	16 January 2025
Amendment 1.4 (EU-specific)	05 February 2025
Amendment 1.5 (EU- and CH-specific)	19 May 2025

This clinical study is to be conducted in compliance with the protocol, with the Regulation (EU) No 536/2014, where applicable, and with the principles of Good Clinical Practice.

Confidentiality Statement

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SPONSOR SIGNATURE PAGE

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Study Number:	PHA022121-C305
Protocol Amendment 1.5:	19 May 2025

This clinical study protocol was subject to critical review and has been approved by the Sponsor.

Signed:_____ Date:_____

[redacted]
Clinical Development Lead,
Executive Director

INVESTIGATOR'S AGREEMENT

Study Title:	A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Orally Administered Deucrictibant Extended-release Tablet for Prophylaxis Against Angioedema Attacks in Adolescents and Adults with Hereditary Angioedema
Study Number:	PHA022121-C305
Protocol Amendment 1.5:	19 May 2025

I have read all pages of this clinical study protocol for which Pharvaris Netherlands B.V. is the Sponsor. I agree that it contains all the information required to conduct this study. I agree to conduct the study as outlined in the protocol and to comply with all the terms and conditions set out therein. I confirm that I will conduct the study in accordance with the International Council for Harmonisation (ICH) Good Clinical Practice (GCP) guidelines (ICH E6) and the provisions of the Helsinki Declaration. I will also ensure that all relevant members of my staff have access to copies of this protocol, the ICH GCP guidelines, and the Helsinki Declaration to enable them to work in accordance with the provisions of these documents.

Signed:_____ Date:_____

Name:_____

<enter name and credentials>
<enter title>
<enter affiliation>
<enter address>
<enter phone number>

STUDY ADMINISTRATIVE STRUCTURE

Table 1: Study Administrative Structure

Investigators:	Names and contact details of participating Investigators are available in the electronic Trial Master File (eTMF)	
Sponsor:	Pharvaris Netherlands B.V. J.H. Oortweg 21 2333 CH Leiden The Netherlands	
Sponsor Representatives:	[redacted] Chief Medical Officer	[redacted] Clinical Development Lead, Executive Director
Medical Monitor:	AMS Advanced Medical Services London, United Kingdom	

1. SYNOPSIS

Name of Sponsor/Company: Pharvaris Netherlands B.V. (Leiden, The Netherlands)		
Name of Investigational Product: Deucrictibant Extended-release (XR) Tablet		
Name of Active Ingredient: Deucrictibant		
Protocol Number: PHA022121-C305 (CHAPTER-3)	**Phase:** 3	**Country:** International
Title of Study: A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of Orally Administered Deucrictibant Extended-release Tablet for Prophylaxis Against Angioedema Attacks in Adolescents and Adults with Hereditary Angioedema		
IND Number: 153097		
EU Trial Number: 2024-516247-62		

Study Objectives:

Primary

- To evaluate the efficacy of deucrictibant 40 mg extended-release (XR) tablet compared with placebo for prophylaxis against angioedema attacks

Secondary

- To characterize the efficacy of deucrictibant 40 mg XR tablet in prophylactic treatment of hereditary angioedema (HAE)
- To evaluate the safety and tolerability of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE
- To evaluate the pharmacokinetics (PK) of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE
- To evaluate the impact on health-related quality of life (HRQoL) of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE

Study Endpoints:

Primary Endpoint: Efficacy

- Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168)

Secondary Endpoints: Efficacy

- Time-normalized number of Investigator-confirmed HAE attacks treated with on-demand medication during the 24-week Treatment Period
- Time-normalized number of Investigator-confirmed moderate or severe HAE attacks during the 24-week Treatment Period
- Time-normalized number of Investigator-confirmed severe HAE attacks during the 24-week Treatment Period

- Proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants achieving ≥70% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants achieving ≥90% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants that are HAE attack-free during the 24-week Treatment Period
- Proportion of time without angioedema symptoms during the 24-week Treatment Period

Secondary Endpoints: Safety

- Treatment-emergent adverse events (TEAEs), including serious adverse events (SAEs), adverse events of special interest (AESIs), and TEAEs leading to study drug discontinuation
- Changes in clinical laboratory tests from baseline
- Changes in vital signs from baseline
- Changes in electrocardiogram (ECG) parameters from baseline

Secondary Endpoint: Pharmacokinetics

- Deucrictibant plasma concentration-time profiles

Secondary Endpoints: Patient Reported Outcomes

- Angioedema Quality of Life (AE-QoL) questionnaire
- Patient Global Assessment of Change (PGA-Change)
- Angioedema Control Test 4-week version (AECT-4wk)
- Work Productivity and Activity Impairment Questionnaire: Specific Health Problem (WPAI-SHP)
- Treatment Satisfaction Questionnaire for Medication (TSQM-9)

Study Design:

This is a Phase 3, multicenter, randomized, double-blind, placebo-controlled study to evaluate the efficacy and safety of orally administered deucrictibant XR tablet once daily (QD) for prophylaxis to prevent attacks of HAE in participants aged ≥12 years.

Following signing of the informed consent form (ICF), participants will undergo screening assessments. The Screening Period will last up to 10 weeks (70 days) to confirm underlying disease status and assess eligibility for study participation. The eligibility of participants receiving HAE medications for long-term prophylaxis (LTP) should be compliant with eligibility criteria.

The baseline attack rate will be based on results from the Run-in Period, which occurs during the Screening Period and will last up to 8 weeks (56 days). The Run-in Period will begin after electronic diary (eDiary) activation during the Screening Visit and will end after 14, 28, 42, or 56 days in case the baseline attack criterion has been met. The signing of the ICF, Screening Visit, and start of Run-in Period should all occur on the same day. During the Run-in Period, the Investigator and/or designee will consult with the participant (or participant's parent/legally designated representative/guardian) weekly to discuss occurrences of HAE attacks. The baseline attack rate requirement for eligibility should not be disclosed to study participants.

Participants must experience [redacted] Investigator-confirmed attacks during the up to 8-week Run-in Period as detailed in the eligibility criteria. Randomization should take place as soon as possible after the baseline attack criterion and all other eligibility criteria have been met.

Participants cannot be randomized during an HAE attack period. An HAE attack must be resolved for at least 24 hours before randomization.

Eligible participants will be randomized in a 2:1 ratio to receive deucrictibant 40 mg XR tablet or matching placebo QD for 24 weeks. Stratification factors of the randomization will be age group (adolescents [≥12 to <18 years] and adults [≥18 years]) and baseline HAE attack rate ([redacted] attacks per 4 weeks versus [redacted] attacks per 4 weeks).

Throughout the study, an eDiary will be completed by participants to record HAE attacks, standard of care on-demand treatments, daily administration of study drug, and to complete electronic patient-reported outcomes (ePROs).

Participants must have access to standard of care on-demand treatments and be able to treat any HAE attack that occurs.

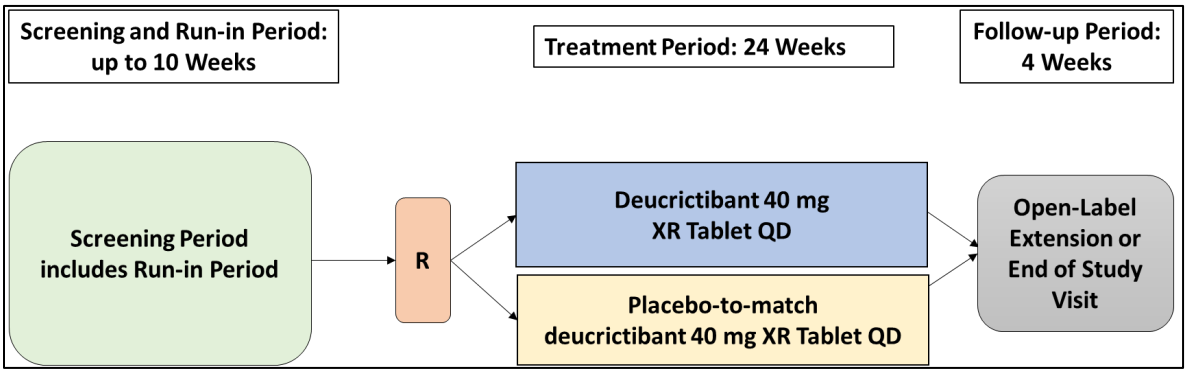
Participants who discontinue study drug before Week 24 of the Treatment Period will complete the End of Treatment (EOT) Visit and will be encouraged to remain in the study per the study protocol defined visit schedule through the End of Study (EOS) Visit.

Participants who withdraw early from the study will be encouraged to complete the EOT and EOS Visits after their last administration of study drug. The assessments for both visits may be done at the same time. For participants who have already discontinued study drug and completed the EOT Visit at an earlier time, only the EOS Visit needs to be completed when they withdraw from the study.

Participants who complete the 24-week Treatment Period will enter a 4-week Follow-up Period or may roll over into the open-label study PHA022121-C307.

Figure 1 provides a schematic of the study design.

Figure 1: Study Design



Abbreviations: QD = once daily; R = randomization (2:1); XR = extended-release.

Study Population:

Adolescent and adult participants (male or female, ≥12 years of age) with HAE will be randomized.

HAE diagnosis will be confirmed by diagnostic testing during the Screening Period. To qualify for randomization, participants must experience [redacted] attacks during the up to 8-week Run-in Period.

The study will be conducted at approximately 80 study sites in approximately 30 countries.

Number of Participants (Planned):

Approximately 81 participants with HAE, including at least 8 adolescents, are planned to be randomized in the study.

Eligibility Criteria:

Inclusion Criteria:

Participants must meet all of the following criteria to be eligible to participate in this study:

1. Provision of written informed consent. At the time of signing informed consent, male and female participants must be aged ≥12 years. Adolescent participants (ie, aged ≥12 to <18

years or as determined by local law) must also be ≥40 kg in body weight at Screening. If the participant is an adolescent, written assent will be obtained from the participant and consent will be obtained from the participant's parent/legally designated representative/guardian, per local regulations. A participant who is an adolescent will sign the ICF if they reach the age of 18 years or as determined by local law during their participation in the study.

2. Participants who previously completed Study PHA022121-C306 may be eligible to be screened for this study. For these participants, there is no need to repeat HAE diagnostic test as described in Inclusion Criterion #3 if the C1 esterase inhibitor (C1INH) function was confirmed by the central laboratory in Study PHA022121-C306.
3. Diagnosis of HAE type 1/2 or type 3 based on the following:
   a. For participants with HAE type 1/2:
      - Documented clinical history consistent with HAE (cutaneous or submucosal, nonpruritic swelling without accompanying urticaria)
      - At least one of the following:
        – Age at reported onset of first angioedema symptoms ≤30 years
        – Family history consistent with HAE
        – C1q above the lower limit of the normal range by central laboratory as part of the Screening assessments
      - Diagnostic testing results to confirm HAE:
        – C1INH functional level <50% of the normal level must be shown by chromogenic assay performed by the central laboratory as part of the Screening procedures. Participants with functional C1INH level 40% to <50% of the normal level must also have a C4 level below the normal range.

   b. For participants with HAE type 3:
      - Recurrent angioedema attacks with diagnostic testing results obtained during Screening to confirm C1INH function ≥50% of normal and C4 level not below the lower level of the normal range performed by the central laboratory
      - Must meet one of the following:
        – Documented genetic mutation associated with HAE type 3 as listed in the Hereditary Angioedema Association (HAEA) and World Allergy Organization (WAO)/European Academy of Allergy and Clinical Immunology (EAACI) Guidelines (Appendix 2)

          Or

        – If no documented genetic mutation:
          - Clinical diagnosis with family history of HAE type 3 and an elevated bradykinin peptide level previously confirmed by a commercially available liquid chromatography-tandem mass spectrometry (LC-MS/MS) assay

            And

- Attacks not responding to treatments with high-dose antihistamine (cetirizine 40 mg/day or equivalent high-dose second-generation antihistamine medication) and no clinical attack symptom relief if treated with corticosteroid, montelukast, or omalizumab

- Documented effective attack symptom relief with on-demand icatibant treatment

Note: The C1INH function sample should be collected at least 5 half-lives after the last dose of C1INH therapy, if applicable.

4. History of at least 3 HAE attacks within the 3 consecutive months prior to Screening Visit

   Note: For participants taking long-term prophylactic HAE therapy before Screening, the at-least 3 attacks may be over the last 3 consecutive months before commencing the prophylactic treatment.

5. Participants must experience [redacted] Investigator-confirmed attacks during the up to 8-week Run-in Period. Randomization should take place after any of the following baseline attack criteria has been met:

   - [redacted]

   Note: The baseline attack rate requirement for eligibility should not be disclosed to study participants.

6. Participant is assessed by the Investigator to have reliable access and ability to use standard of care on-demand treatments to effectively manage acute HAE attacks.
7. Investigator considers that the participant (and parent/legally designated representative/guardian for adolescent participants) is willing and able to adhere to all protocol requirements, including ensuring that the participant is capable of, and compliant with, eDiary and ePRO data recording.
8. Female participants of childbearing potential (or who become of childbearing potential during the study) must agree to the protocol specified pregnancy testing and to be abstinent from heterosexual intercourse or to use a highly effective contraception method as defined in the protocol (Section 8.2.2) and as available locally, from enrollment until 30 days after the last study drug administration.

   There are no contraceptive requirements for male participants.

Exclusion Criteria:

Participants who meet any of the following criteria will be excluded from the study.

1. Any diagnosis of angioedema other than HAE
2. Participation in a clinical study with any other investigational drug within the last 30 days or within 5 half-lives of the investigational drug at Screening (whichever was longer)
3. Has received prior prophylactic treatment or on-demand treatment with deucrictibant, with the exception of participants from Study PHA022121-C306.

4. Exposure to angiotensin-converting enzyme (ACE) inhibitors or any estrogen-containing medications with systemic absorption (such as oral contraceptives or hormonal replacement therapy) within 4 weeks of Screening
5. Prior gene therapy for any indication at any time
6. Receiving prophylactic treatment for HAE and are satisfied with this treatment. Patients who are not satisfied (eg, tolerability issues, lack of efficacy) and have previously stopped long-term prophylactic HAE treatment can sign the ICF and begin the Screening Period only if their last dose of treatment was received prior to the time point before Screening indicated below:
   a. Long-term prophylactic therapy for HAE with C1INH, oral kallikrein inhibitors, or anti-fibrinolytics: 2 weeks prior to Screening
   b. Long-term prophylactic therapy for HAE with attenuated androgens: 4 weeks prior to Screening
   c. Long-term prophylactic monoclonal antibody therapy for HAE (ie, lanadelumab): 5 half-lives prior to Screening
   d. Short-term prophylaxis for HAE: 7 days prior to Screening. Note: Short-term prophylaxis is defined as intravenous (iv) C1INH to avoid angioedema complications from medically indicated procedures.

   Note: If prophylaxis is not listed above, please consult with Sponsor.
7. Any females who are pregnant, plan to become pregnant, or are currently breast-feeding
8. Abnormal hepatic function (aspartate aminotransferase [AST] >2× upper limit of normal [ULN], alanine aminotransferase [ALT] >2× ULN, or total bilirubin >1.5× ULN or any hepatic impairment via Child-Pugh Scoring System). Participants with Gilbert's syndrome, defined as isolated increase of total bilirubin ≤3× ULN and AST and ALT within the normal range, are not excluded.
9. Moderate or severe renal impairment (estimated glomerular filtration rate [eGFR] <60 mL/min/1.73 m^2)
10. Any clinically significant history of angina, myocardial infarction, syncope, stroke, left ventricular hypertrophy or cardiomyopathy, uncontrolled hypertension, bradycardia, or any other clinically significant cardiovascular abnormality within the previous year that, in the opinion of the Investigator, would interfere with the participant's safety or ability to participate in the study
11. History of epilepsy and/or other significant neurological diseases
12. Any clinically significant and uncontrolled gastrointestinal dysfunction (eg, chronic diarrhea, inflammatory bowel disease) which impacts study drug absorption
13. History of alcohol or drug abuse within the previous year, or current evidence of substance dependence or abuse
14. Use of concomitant medications with systemic absorption and foods that are moderate and strong inhibitors of cytochrome P450 (CYP) 3A4 such as clarithromycin, erythromycin, diltiazem, itraconazole, ketoconazole, ritonavir, verapamil, and grapefruit juice or strong inducers of CYP3A4 such as carbamazepine, phenytoin, and rifampin within the last 30 days or within 5 half-lives (whichever is longer) of the time of randomization
15. Known hypersensitivity to deucrictibant or any of the excipients of the study drug

Study Drug, Dosage, and Mode of Administration:

Deucrictibant has been adopted as the international nonproprietary name (INN) for the active ingredient, previously referred to as PHA121 or PHA-022121. Deucrictibant 40 mg XR is a tablet formulation.

The study drug consists of deucrictibant 40 mg XR tablet and matching placebo tablet for oral use. The QD administration of study drug throughout the Treatment Period is recommended to occur at the same time daily without a meal or with a light meal, with 24 ± 2 hours between each dose when study drug is not administered during a study visit. The tablets are to be swallowed with water.

Duration of Treatment:

24 weeks

Management of HAE Attacks:

Participants must have access to standard of care on-demand treatments and be able to treat any HAE attack that occurs. Participants should continue administering study drug and following study procedures without alteration to the protocol schedule, even if they receive standard of care on-demand treatment for HAE attacks. All HAE medications associated with the management of attacks will be reported.

Criteria for Evaluation:

Efficacy

Efficacy will be assessed based upon the occurrences of HAE attacks, which will be confirmed by Investigators. During the Screening Visit, site personnel will discuss with participants the requirements for reporting HAE attacks, including the information that is to be captured for each HAE attack.

Participants will receive training on how to use the eDiary at the Screening Visit and throughout the study, as needed. Information on HAE attacks will be collected in an eDiary by the participant. Each reported HAE attack will be assessed by the Investigator and/or designee and confirmed by the Investigator within 72 hours from onset of symptoms of an attack.

The criteria provided in this section should be used by the Investigator to assess if an event reported by a participant can be confirmed as an HAE attack.

To be confirmed as an HAE attack, the event must have signs or symptoms consistent with an attack in at least one of the following locations:

- Peripheral angioedema: cutaneous swelling involving an extremity, face, neck, torso and/or genitourinary region
- Abdominal angioedema: abdominal pain, with or without abdominal distension, nausea, vomiting, or diarrhea
- Upper airway angioedema: sensation of lump in the throat, difficulty swallowing, throat tightening, difficulty speaking, hoarseness, stridor, dyspnea, or upper airways swelling on the tongue, palate, uvula, or larynx

Despite the presence of these symptoms, the Investigator may still determine that the event does not represent an HAE attack if there are clinical features that strongly refute such a diagnosis (eg, associated with wheals or longer in duration than the typical time course of an attack or alternate etiology such as recent trauma resulting in swelling or pain).

To be counted as a unique HAE attack distinct from a previous attack, the new symptoms must occur at least 24 hours after complete resolution of symptoms of a prior attack (ie, there must be a 24-hour symptom-free period to be considered a new attack).

Participant-reported occurrences of prodromal symptoms alone or pre-emptive use of acute treatments without swelling symptoms are not considered to meet the criteria of an HAE attack.

The following information will be entered by the participant in the eDiary:

- Date and time of onset of symptoms of an HAE attack
- Date and time of complete resolution of attack symptoms
- If standard of care on-demand treatment was administered for the HAE attack

Before the Investigator and/or designee completes a separate Investigator Assessment Form for each participant-reported HAE attack, they will review the data entered by the participant via the eDiary vendor portal and follow-up with the participant. Follow-up topics include, but are not limited to:

- Location of attack symptoms
- Start/end date/time of symptoms
- Attack severity (mild, moderate, or severe)
- If standard of care on-demand medications were administered for the HAE attack (date, time, and medication(s) including the number of doses administered)
- Type of medical assistance or intervention, if required
- If hospitalization or a visit to emergency department occurred

Additional queries by the Investigator and/or designee during participant follow-up may include questions such as: whether the participant only experienced prodromal symptoms, whether the symptoms experienced were typical or different from prior attacks, or whether there were possible alternative etiologies for the symptoms.

The outcome of the follow-up will determine if an HAE attack is confirmed.

Patient Reported Outcomes

AE-QoL, PGA-Change, AECT-4wk, WPAI-SHP, and TSQM-9 will be used to assess health-related QoL, disease control, work productivity, and treatment satisfaction, respectively.

Safety

Safety and tolerability will be assessed based on the following:

- Adverse events
- Safety laboratory testing (hematology, biochemistry, coagulation, urinalysis, and pregnancy test)
- Physical exams and vital signs (heart rate, blood pressure, and body temperature)
- 12-lead ECG

Pharmacokinetics

Pharmacokinetic (PK) parameters based on plasma concentration of deucrictibant will be assessed.

Statistical Methods:

A Statistical Analysis Plan (SAP) will be finalized prior to database lock and will include a more technical and detailed description of all planned statistical analyses of all study endpoints.

A Poisson distribution of Investigator-confirmed HAE attacks was assumed for the analysis of primary endpoint 'Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment period'. In the Phase 2 study PHA022121-C301, [redacted] in the placebo group. With a conservative assumption that the time-normalized number of Investigator-confirmed HAE attacks [redacted] in the placebo group during the 24-week Treatment Period, a sample size of 81 randomized participants (2:1 randomization; 54 on deucrictibant 40 mg XR tablet and 27 on placebo) will provide a power of at least 95% to detect a 50% reduction in time-normalized number of Investigator-confirmed HAE attacks for deucrictibant 40 mg XR comparing to placebo, based on a two-sided test at a significance level of 5%.

The primary endpoint is the time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168).

The primary estimand of interest is defined by the following attributes:

- Treatment: Deucrictibant 40 mg XR tablet versus placebo, orally administered QD
- Population: The target patient population is defined through the study inclusion/ exclusion criteria.
- Endpoint: Time normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168). Unless otherwise specified, this is also referred to as the 4-week HAE attack rate, or HAE attack rate, which is calculated for each participant as the number of Investigator-confirmed HAE attacks occurring during the 24-week Treatment Period divided by the number of days the participant contributed to the 24-week Treatment Period, multiplied by 28 days.
- Intercurrent events (ICEs):
  - Early study drug discontinuation for any reason unless accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period
  - Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period
  - Use of prohibited medication (including prohibited short/long-term HAE prophylaxis) during the 24-week Treatment Period
  - Non-compliance to study drug during the 24-week Treatment Period
- Population-level summary: Ratio of the 4-week HAE attack rates between deucrictibant 40 mg XR tablet and placebo

Composite variable strategy will be used to address the ICE 'Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period'. For the remaining ICEs, a treatment policy strategy will be applied.
The analysis of the primary efficacy endpoint will be performed based on the Intent-to-Treat (ITT) Analysis Set and will compare deucrictibant 40 mg XR tablet with placebo using a generalized linear model (GLM) assuming a Poisson distribution with a log link function and Pearson chi-square scaling of standard errors to account for potential (over)-dispersion. The model will include fixed effects for treatment group, the baseline attack rate and age group (adolescent and adult), and an offset term of the logarithm of participant's follow-up time during the Treatment Period. The estimated least squares (LS) rate for deucrictibant 40 mg XR tablet and placebo and the rate ratio relative to placebo will be provided with 95% confidence intervals. These estimates will be reported as 4-week attack rates by transforming the estimates using the exponential function and scaling by the unit of time. The

treatment effect in the primary efficacy endpoint evaluation will be tested using a Wald-based chi-square test.

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3. LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

The following abbreviations and specialist terms are used in this study protocol.

Table 2: Abbreviations and Specialist Terms

Abbreviation	Definition
AECT-4wk	Angioedema Control Test 4-week version
ADL	activities of daily living
AE	adverse event
AE-QoL	Angioedema Quality of Life questionnaire
AESI	adverse event of special interest
ALT	alanine aminotransferase
ANGPT1	angiopoietin-1
AST	aspartate aminotransferase
BID	twice daily
BK	bradykinin
C1INH	C1 esterase inhibitor
CFR	Code of Federal Regulations
C_{max}	peak plasma concentration
CNS	central nervous system
CRO	contract research organization
CTCAE	Common Terminology Criteria for Adverse Events
CYP	cytochrome P450
ECG	electrocardiogram
eCRF	electronic case report form
eTMF	electronic trial master file
EAACI	European Academy of Allergy and Clinical Immunology
EDC	electronic data capture
EFD	embryofetal development
EMA	European Medicines Agency
EOS	end of study
EOT	end of treatment
ePRO	electronic patient-reported outcome

Abbreviation	Definition
EU	European Union
FDA	Food and Drug Administration
FSH	follicle-stimulating hormone
FXII	Factor XII
GCP	Good Clinical Practice
GDPR	General Data Protection Regulation
GLM	general linear model
HAE	hereditary angioedema
HAEA	Hereditary Angioedema Association
hCG	human chorionic gonadotropin
HRQoL	health-related quality of life
ICE	intercurrent event
ICF	informed consent form
ICH	International Conference on Harmonisation
IDMC	Independent Data Monitoring Committee
IEC	Independent Ethics Committee
IRB	Institutional Review Board
IRT	interactive response technology
ITT	Intent-to-Treat
iv	intravenous(ly)
KNG1	kininogen-1
LC-MS/MS	liquid chromatography-tandem mass spectrometry
LS	least squares (mean)
LTP	long-term prophylaxis
MedDRA	Medical Dictionary for Regulatory Activities
MYOF	myoferlin
NCI	National Cancer Institute
PGA-change	Patient Global Assessment of Change
PK	pharmacokinetic(s)
PLG	plasminogen

Abbreviation	Definition
PT	preferred term
QA	quality assurance
QD	once daily
QoL	quality of life
SAE	serious adverse event
SAP	Statistical Analysis Plan
sc	subcutaneous(ly)
SOC	system organ class
SOP	standard operating procedure
SUSAR	suspected unexpected serious adverse reaction
TEAE	treatment-emergent adverse event
TSQM	Treatment Satisfaction Questionnaire for Medication
WAO	World Allergy Organization
WHO	World Health Organization
WPAI-SHP	Work Productivity and Activity Impairment Questionnaire: Specific Health Problem
XR	extended-release

4. INTRODUCTION

4.1. Hereditary Angioedema

Hereditary angioedema (HAE) is a rare disease that manifests as unpredictable and recurrent edema attacks of cutaneous or submucosal tissue most commonly affecting the limbs, intestinal tract, face (lips and tongue), genitals, and airways. HAE attacks often lead to discomfort and pain, and can include nausea, vomiting and diarrhea when the intestinal tract is affected. HAE attacks can become life-threatening in the case of airway obstruction and asphyxiation if the attack remains untreated. The number, duration, and severity of attacks vary highly between patients and throughout a patient's lifetime. Trauma, stress, and medical procedures are known triggers for attacks.

Three types of HAE have been characterized. HAE type 1 and type 2 (HAE type 1/2) are defined by mutations in the C1 esterase inhibitor (C1INH) gene and result either in reduced levels (type 1) or reduced activity (type 2) of C1INH. In both cases, decreased C1INH protease activity leads to uncontrolled activity of Factor XII and plasma kallikrein, resulting in an activated kallikrein-kinin system, ie, excessive cleavage of high molecular weight kininogen and increased plasma levels of bradykinin (BK).

HAE with normal C1INH (type 3) is characterized by similar clinical manifestations to HAE type 1/2 in the presence of normal levels and activity of C1INH and, in a number of cases, is associated with characterized mutations in other genes associated with HAE type 3 as listed in the Hereditary Angioedema Association (HAEA) and World Allergy Organization (WAO)/European Academy of Allergy and Clinical Immunology (EAACI) Guidelines (Appendix 2). Recently, some laboratory tests have been developed to identify these patients by measuring the BK levels in the blood and stimulated kallikrein activity (Li, 2023, Lara-Marquez, 2018).

BK is the principal mediator of the signs and symptoms associated with HAE attacks. Two BK receptor types, bradykinin B1 and bradykinin B2, have been identified, cloned, and characterized. BK has a high affinity and high agonist potency at the bradykinin B2 receptor, but a near 1000-fold lower affinity and potency at the B1 receptor, making it a selective bradykinin B2 receptor agonist.

The bradykinin B2 receptor is expressed on the cell surface of the luminal side of vascular endothelial cells. Excessive BK generation, as in HAE, promotes vascular permeability, leading to plasma extravasation and cutaneous or submucosal tissue edema and swelling typical of an angioedema attack.

4.2. Current Treatment Options for Hereditary Angioedema

There are currently 2 treatment approaches to the management of HAE: acute (on-demand) treatment of attacks and prevention of attacks with short- or long-term prophylactic therapy. Patients and their physicians have several choices for on-demand or prophylactic treatment, but because there are differences in the route of administration, side effect profiles, and efficacy between medications, patient preference and experience are considered in selecting the most appropriate therapy.

Patients must have access to standard of care on-demand HAE medication that can be self-administered. Treatment should be taken regardless of the location of the attack, and treatment should be initiated early to mitigate progression of the attack. Regular monitoring of the patient's frequency of use of on-demand treatment and treatment outcomes should be conducted (Busse, 2021, Maurer, 2022). In addition to on-demand treatment of HAE attacks, a patient may also require prophylactic treatment.

The requirement for long-term prophylactic treatment is not made on rigid criteria, but rather the needs of the individual patient (Busse, 2021, Maurer, 2022). The decision to initiate long-term prophylaxis is made with consideration for the patient's quality of life and treatment preferences in the context of attack frequency, attack severity, comorbid conditions, and access to treatment (Busse, 2021, Maurer, 2022). Because disease severity may change over time, the need to initiate or continue long-term prophylactic treatment is periodically reviewed.

4.2.1. Prophylactic Treatments

The plasma-derived C1INH concentrates (Cinryze® and Haegarda®/Berinert® 2,000/3,000) are administered by intravenous (iv) and subcutaneous (sc) routes, respectively. These products are administered twice weekly (every 3 or 4 days). Plasma kallikrein inhibitors include lanadelumab (Takhzyro®), a fully human recombinant monoclonal antibody, which is administered sc every 2 or 4 weeks, and berotralstat (Orladeyo®), which is administered orally, once daily. Attenuated androgens (including danazol, stanozolol, oxandrolone, oxymetholone, tibolone, methyltestosterone) and antifibrinolytics, such as ε-aminocaproic acid and tranexamic acid, are used for the prevention of acute attacks and are licensed in some countries. However, current treatment guidelines recommend against the use of attenuated androgens and antifibrinolytics as first-line therapy due to limited efficacy and/or safety issues with associated contraindications (Busse, 2021, Maurer, 2022).

4.3. Unmet Medical Needs in Managing Bradykinin-Mediated Angioedema

There is a high unmet medical need for efficacious oral medicines that can facilitate timely treatment of attacks due to HAE and other various forms of bradykinin-mediated angioedema and thereby improve outcomes, compliance, and the overall care and quality of life for patients. All but one of the medications approved in the last decade for the management of HAE are administered parenterally via iv or sc routes, which are often associated with a delay of drug administration, injection pain, local side effects, and a lower quality of life. In a 2017 Food and Drug Administration (FDA) Patient-Focused Drug Development meeting, patients expressed that the route of administration (iv, sc, or oral) is the most important consideration for a new HAE therapeutic, with a preference for oral treatments (FDA, 2018).

4.4. Deucrictibant

Deucrictibant is under development as an orally administered BK B2 receptor antagonist with potential for both acute therapeutic and prophylactic use in patients with different types of BK-mediated angioedema, including HAE type 1/2, HAE with normal C1INH (type 3), and acquired angioedema due to C1INH deficiency.

Deucrictibant has been extensively characterized in primary pharmacological studies as a potent, orally available B2 receptor antagonist.

Deucrictibant did not interact in vitro with human cardiac ion channels known to contribute to the cardiac action potential, including the human ether-a-go-go-related gene, at up to concentrations 211- to 228-fold higher than exposure levels anticipated in this study. Nor did single oral doses of deucrictibant induce any relevant changes in heart rate, arterial blood pressure, or body temperature, or in any electrocardiogram (ECG) parameters assessed in the cynomolgus monkey. In addition, there were no clinically significant treatment-related adverse effects of deucrictibant observed in ECG parameters, cardiac weights, or microscopic evaluation of cardiac tissue in the repeated dose toxicology studies or 28-day tolerability study.

Completed Phase 1 clinical studies did not reveal any deucrictibant-related cardiac disorders. Furthermore, no cardiac disorders were reported in the Phase 2 HAE studies PHA022121-C201 or PHA022121-C301. There were no clinically significant changes in ECG parameters observed across all single- and multiple-dose clinical studies. A concentration-QTcF analysis indicated that deucrictibant does not demonstrate QT prolonging effects at concentrations that cover the therapeutic dose range as well as the high clinical exposure scenarios for both on-demand and prophylactic treatment.

In nonclinical toxicology studies, acute adverse central nervous system (CNS)-related clinical signs at the time of peak plasma concentration (C_{max}) were observed in several animal species on Day 1 or 2 after administration of supratherapeutic doses of deucrictibant at systemic exposure 20- to 90-fold higher than the anticipated unbound plasma C_{max} in humans administered deucrictibant 40 mg XR tablet.

In the recently completed 26-week video-electroencephalography (EEG) study in rats, no increased risk was attributed to deucrictibant compared to placebo after long-term administration.

As of the clinical safety cutoff date of 10 June 2024, all events except for 1 headache categorized under the System Organ Class, "Nervous System Disorder" were mild or moderate in intensity and the incidence rate has been comparable between active and placebo groups. One severe headache was reported and considered by the Investigator to be unrelated to study drug.

Deucrictibant had no adverse effects on male or female fertility or embryofetal development (EFD) in the humanized bradykinin B2-receptor transgenic rat nor on EFD in rabbits, up to the highest dose tested. This supports the inclusion of women of childbearing potential in the Phase 3 clinical studies provided appropriate contraception precautions are in place. Functional assessment of male fertility has been completed both in the wild-type rat as part of the 6-month repeat-dose toxicity study and in a dedicated male fertility study in the humanized bradykinin B2-receptor transgenic rat following pre-mating dosing of at least 10 weeks. There are no effects on male reproductive parameters up to the highest dose tested. In addition, there were no organ weight or histopathological changes of concern in the rat reproductive tract of either sex. In conjunction with the EFD results, the clinical studies do not require male participants to use male contraception.

Overall, in completed Phase 1 studies (using either deucrictibant oral solution or soft capsule) and in the Phase 2 clinical study of deucrictibant prophylactic treatment to prevent attacks of HAE in adults with HAE (PHA022121-C301), deucrictibant has been shown to be generally

well-tolerated. Most treatment-emergent adverse events (TEAEs) reported have been mild to moderate in severity. No severe TEAEs or serious adverse events (SAEs) have been considered by the Investigator as related to deucrictibant. Additional safety information is provided in the Investigator's Brochure.

In the Phase 2 Study PHA022121-C301, results from the double-blinded Part 1 demonstrated that treatment with deucrictibant 10 mg soft capsule twice daily (BID) and 20 mg soft capsule BID statistically significantly reduced the monthly HAE attack rate by 79.3% and 84.5% compared to placebo, respectively, as well as markedly reduced moderate or severe attacks (82.8% and 92.3%) and attacks treated with on-demand HAE medications (75.1% and 92.6%). In addition, overall, deucrictibant was generally safe and well-tolerated, with no reported treatment-emergent SAEs or TEAEs leading to treatment discontinuation. TEAEs reported were generally balanced among each treatment group. No clinically significant abnormal trends of laboratory parameters, vital signs, or ECG parameters were observed.

4.4.1. Benefit-Risk with Deucrictibant

Given the potential benefit to treat the HAE population, especially with the validated mechanism of action, positive pharmacodynamic and clinical effects demonstrated by the BK challenge in Study PHA022121-C001, and proof of concept to prevent HAE attacks in Study PHA022121-C301, the benefit-risk assessment of deucrictibant is considered acceptable if used according to the clinical study protocol, and in accordance with the recommendations and guidance provided in the Investigator's Brochure.

Based on nonclinical and clinical data to date, the cardiovascular risk of deucrictibant is considered low. However, as a precaution, participants with clinically significant cardiovascular history will not be eligible to enroll in the study and cardiovascular health will be monitored with 12-lead ECGs, vital signs, and physical examinations throughout the study.

While the neurological effects in nonclinical studies are believed to be associated with supratherapeutic C_{max} systemic concentrations, neurological AEs will be monitored as adverse events of special interest (AESIs) in this study as a precautionary measure, and participants with epilepsy or other significant neurological diseases will not be eligible to enroll in the study.

Hereditary angioedema is a disease with the same pathophysiology in adults and pediatric patients. There is a lack of evidence demonstrating any significant differences in HAE etiology and pathophysiology based on age, and the overall clinical aspects of adult and pediatric HAE are quite similar (Farkas, 2017, Wahn, 2020). The comparable exposure-response relationship is thus anticipated in view of the development biology of the kallikrein-kinin system. Moreover, the adult and pediatric data from icatibant, sharing the same mechanism of action, supports this assumption (Farkas, 2017, Wang, 2021, EMA, 2017).

Overall, the benefit-risk assessment appears to be favorable to support the clinical investigation of deucrictibant for prophylactic treatment to prevent HAE attacks in the adult and pediatric populations planned for enrollment in this study.

Refer to the Investigator's Brochure for additional details regarding the benefit-risk with deucrictibant.

5. STUDY OBJECTIVES AND ENDPOINTS

The study objectives and endpoints are listed in Table 3.

Table 3: Objectives and Endpoints

Objectives	Endpoints
Primary	
• To evaluate the efficacy of deucrictibant extended-release 40 mg (XR) tablet compared with placebo for prophylaxis against angioedema attacks	• Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168)
Secondary	
• To characterize the efficacy of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE	• Time-normalized number of Investigator-confirmed HAE attacks treated with on-demand medication during the 24-week Treatment Period • Time-normalized number of Investigator-confirmed moderate or severe HAE attacks during the 24-week Treatment Period • Time-normalized number of Investigator-confirmed severe HAE attacks during the 24-week Treatment Period • Proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period • Proportion of participants achieving ≥70% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period • Proportion of participants achieving ≥90% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period • Proportion of participants that are HAE attack-free during the 24-week Treatment Period • Proportion of time without angioedema symptoms during the 24-week Treatment Period
• To evaluate the safety and tolerability of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE	• Treatment-emergent adverse events (TEAEs), including serious adverse events (SAEs), adverse events of special interest (AESIs), and TEAEs leading to study drug discontinuation • Changes in clinical laboratory tests from baseline • Changes in vital signs from baseline • Changes in electrocardiogram (ECG) parameters from baseline

Objectives	Endpoints
• To evaluate the pharmacokinetics (PK) of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE	• Deucrictibant plasma concentration-time profiles
• To evaluate the impact on health-related quality of life (HRQoL) of deucrictibant 40 mg XR tablet in prophylactic treatment of HAE	• Angioedema Quality of Life (AE-QoL) questionnaire • Patient Global Assessment of Change (PGA-Change) • Angioedema Control Test 4-week version (AECT-4wk) • Work Productivity and Activity Impairment Questionnaire: Specific Health Problem (WPAI-SHP) • Treatment Satisfaction Questionnaire for Medication (TSQM-9)

6. INVESTIGATIONAL PLAN

6.1. Overall Study Design

This is a Phase 3, multicenter, randomized, double-blind, placebo-controlled study to evaluate the efficacy and safety of orally administered deucrictibant XR tablet once daily (QD) for prophylaxis to prevent attacks of HAE in participants aged ≥12 years.

Following signing of the informed consent form (ICF), participants will undergo screening assessments. The Screening Period will last up to 10 weeks (70 days) to confirm underlying disease status and assess eligibility for study participation. The eligibility of participants receiving HAE medications for long-term prophylaxis (LTP) will be compliant with eligibility criteria.

The baseline attack rate will be based on results from the Run-in Period, which occurs during the Screening Period and will last up to 8 weeks (56 days). The Run-in Period will begin after electronic diary (eDiary) activation during the Screening Visit and will end after 14, 28, 42, or 56 days in case the baseline attack criterion has been met. The signing of the ICF, Screening Visit, and start of Run-in Period should all occur on the same day. During the Run-in Period, the Investigator and/or designee will consult with the participant (or participant's parent/legally designated representative/guardian) weekly to discuss occurrences of HAE attacks. The baseline attack rate requirement for eligibility should not be disclosed to study participants.

Participants must experience [redacted] Investigator-confirmed attacks during the up to 8-week Run-in Period as detailed in the eligibility criteria. Randomization should take place as soon as possible after the baseline attack criterion and all other eligibility criteria have been met.

Participants cannot be randomized during an HAE attack period. An HAE attack must be resolved for at least 24 hours before randomization.

Eligible participants will be randomized in a 2:1 ratio to receive deucrictibant 40 mg XR tablet or matching placebo QD for 24 weeks. Stratification factors of the randomization will be age group

(adolescents [≥12 to <18 years] and adults [≥18 years]) and baseline HAE attack rate ([redacted] per 4 weeks versus [redacted] attacks per 4 weeks).

Throughout the study, an eDiary will be completed to record HAE attacks, standard of care on-demand treatments, daily administration of study drug, and to complete electronic patient-reported outcomes (ePROs).

Participants must have access to standard of care on-demand treatments and be able to treat any HAE attack that occurs.

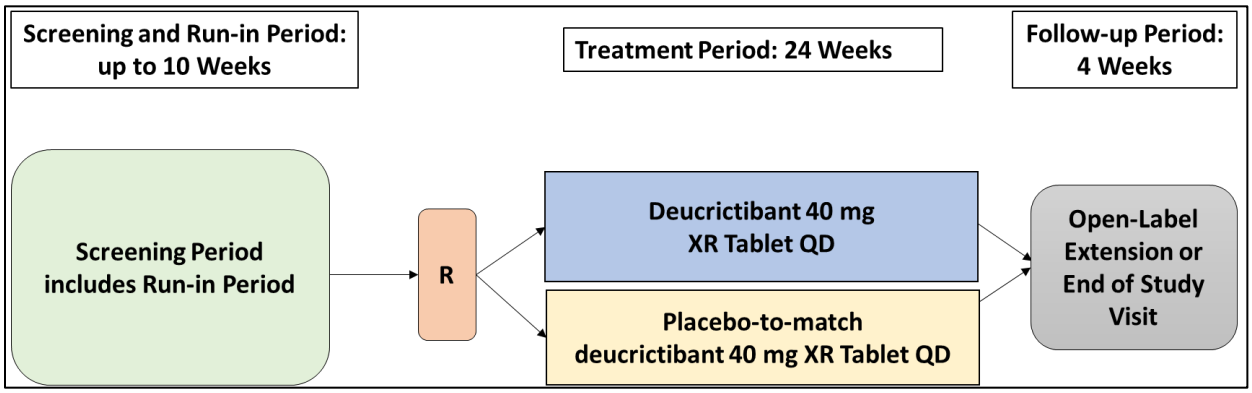
Participants who discontinue study drug before Week 24 of the Treatment Period will complete the End of Treatment (EOT) Visit and will be encouraged to remain in the study per study protocol defined visit schedule through the End of Study (EOS) Visit.

Participants who withdraw early from the study will be encouraged to complete the EOT and EOS Visits after their last administration of study drug. The assessments for both visits may be done at the same time. For participants who have already discontinued study drug and completed the EOT Visit at an earlier time, only the EOS Visit needs to be completed when they withdraw from the study.

Participants who complete the 24-week Treatment Period will enter a 4-week Follow-up Period or may roll over into the open-label Study PHA022121-C307.

Figure 2 provides a schematic of the study design.

Figure 2: Study Design



Abbreviations: QD = once daily; R = randomization (2:1); XR = extended-release.

6.1.1. Schedule of Events

The Schedule of Events is presented in Table 4.

Table 4: Schedule of Events

Visit (Week/Day)	Screening and Run-in Period [a]		Treatment Period (24 weeks)							Follow-up Period
	Screening Visit [b]	Run-in Period [b, c]	Visit 1 Day 1	Visit 2 [d] Week 2 Day 15	Visit 3 [c] Week 4 Day 29	Visit 4 Week 8 Day 57	Visit 5 [d] Week 16 Day 113	Visit 6 [e] Week 24 Day 168	EOT Visit [e]	EOS Visit [c, e]
Visit Window	**NA**	**NA**	**NA**	**± 3 days**	**± 3 days**	**± 5 days**	**± 7 days**	**± 7 days**	**NA**	
	On-site	**Virtual**	**On-site**	**On-site or Remote**	**Virtual**	**On-site**	**On-site or Remote**	**On-site**	**On-site**	**On-site or Virtual**
Informed consent [b, f]	X									
Contact participant		X [g]			X					
Randomization [h]			X							
Demographics and HAE characteristics [i]	X									
Medical history	X									
Weight and height [j]	X							X	X	
Physical examination [k]	X		X			X		X	X	
Vital signs [l]	X		X	X		X	X	X	X	
12-Lead ECG [m]	X		X			X		X	X	
Clinical laboratory testing [n]	X		X	X		X		X	X	
Optional biomarker sampling	X							X	X	
PK sampling			Refer to Table 5							
Pregnancy test [o]	X		X			X	X	X	X	

Visit (Week/Day)	Screening and Run-in Period [a]		Treatment Period (24 weeks)							Follow-up Period
	Screening Visit [b]	Run-in Period [b, c]	Visit 1 Day 1	Visit 2 [d] Week 2 Day 15	Visit 3 [c] Week 4 Day 29	Visit 4 Week 8 Day 57	Visit 5 [d] Week 16 Day 113	Visit 6 [e] Week 24 Day 168	EOT Visit [e]	EOS Visit [c, e]
Visit Window	**NA**	**NA**	**NA**	**± 3 days**	**± 3 days**	**± 5 days**	**± 7 days**	**± 7 days**	**NA**	
	On-site	**Virtual**	**On-site**	**On-site or Remote**	**Virtual**	**On-site**	**On-site or Remote**	**On-site**	**On-site**	**On-site or Virtual**
FSH test [p]	X									
Adverse events [q]	←									→
Prior / concomitant medications	←									→
Dispense study drug [r]			X	X		X	X			
Dispense standard of care on-demand medication [r]	X (if needed)		X (if needed)	X (if needed)		X (if needed)	X (if needed)			
Study drug administration [s]			←					→		
eDiary and ePRO training [t]	X									
eDiary [u]	←									→
ePRO: AE-QoL questionnaire [v]			X [w]		X		X	X	X	
ePRO: PGA-Change [v]								X	X	
ePRO: AECT-4wk [v]			X [w]		X		X	X	X	
ePRO: WPAI-SHP [v]	X				X			X	X	
ePRO: TSQM-9 [v]	X				X			X	X	

Abbreviations: AECT-4wk = angioedema control test-4 week; AE-QoL = angioedema quality of life questionnaire; ECG=electrocardiogram; eCRF = electronic case report form; EOS = End of Study; EOT = End of Treatment; ePRO=electronic patient report outcome (instrument); HAE = hereditary angioedema;

hCG = human chorionic gonadotropin; ICF = informed consent form; NA = not applicable; PGA-Change = patient global assessment of change question; PK = pharmacokinetic; TSQM-9 = Treatment Satisfaction Questionnaire for Medication; WPAI-SHP = Work Productivity and Activity Impairment Questionnaire: Specific Health Pro.

a The Screening Period begins at the time of signing the ICF and will last up to 10 weeks (70 days) to confirm underlying disease status, assess eligibility for study participation, and to determine the baseline attack rate. The baseline attack rate will be based on results from the Run-in Period, which will last up to 8 weeks (56 days).
Participants must experience [redacted] during the up to 8-week Run-In Period as detailed in the eligibility criteria.

b The Run-in Period will begin after eDiary activation during the Screening Visit and will end after 14, 28, 42, or 56 days in case the baseline attack criterion has been met. The signing of the ICF, Screening Visit, and start of Run-in Period should all occur on the same day.

c The Run-in Period and Visit 3 are virtual visits and the EOS Visit may be virtual. Site personnel (Investigator or designee) will contact participants to ensure the visit procedures are performed.

d Visit 2 and Visit 5 may be performed either on-site or remotely by a visiting nurse or home healthcare provider (where applicable and permitted by local regulations). For Visit 2 and Visit 5, if performed remotely, supply of study drug and standard of care on-demand treatment (if needed) may be provided by a visiting nurse or home healthcare provider, or by direct shipment to the participant's home (where permitted by local regulations).

e For participants who complete the 24-week Treatment Period and are entering the open-label Study PHA022121-C307 on the same day, Visit 6 serves as the EOT and EOS Visits.
For participants who complete the 24-week Treatment Period but are not entering the open-label Study PHA022121-C307, Visit 6 serves as the EOT Visit. These participants will proceed- to the Follow-up Period and EOS Visit. The EOS Visit occurs 14 to 28 days after Visit 6.
The EOT Visit assessments will be done for participants who do not complete the 24-week Treatment Period at the time of study drug discontinuation.
The Follow-up Period is only for participants who are not entering the open-label Study PHA022121-C307.

f At the time of signing the ICF, male and female participants must be aged ≥12 years. If the participant is an adolescent (ie, aged ≥12 to <18 years or as determined by local law), written assent will be obtained from the participant and consent will be obtained from the participant's parent/legally designated representative/guardian, per local regulations. A participant who is an adolescent will sign the informed consent form if they reach the age of 18 or as determined by local law during their participation in the study.

g During the Run-in Period, the Investigator and/or designee will consult with the participant (or participant's parent/legally designated representative/ guardian) weekly to discuss occurrences of HAE attacks. The contacts are expected to occur on Days 7, 21, 35, and 49 (± 2 days) and Days 14, 28, 42, and 56 (+ 2 days) (ie, on or after the time points for the Run-in Period attack rate eligibility assessment).

h Randomization should take place as soon as possible after the baseline attack criterion and all other eligibility criteria have been met.

i Including sampling for laboratory assessment of C1INH function, C4 level, and C1q level (if applicable).

j Height is measured at the Screening Visit for all participants. At Visit 6 (or the EOT Visit, if the participant discontinues study drug early), height will also be measured for participants who were aged ≥12 to <18 years at the time of signing informed consent.

k Physical examination at minimum includes the following body systems: gastrointestinal system; head/eyes/ears/nose/throat; endocrine (including thyroid); respiratory system; cardiovascular system; integumentary system (skin); musculoskeletal; and nervous system (central and peripheral). Additional physical examinations may be scheduled at the discretion of the Investigator (or designee). If any abnormalities in these systems are identified, further examinations or external consultations are required. Clinically significant findings will be recorded on the eCRF.

l Blood pressure, heart rate, and body temperature.

m ECGs (in triplicate) are to be performed at the study visits indicated. ECG recordings should occur prior to the PK sample collection, if applicable.

n Clinical safety laboratory testing includes hematology, chemistry, coagulation, and macroscopic dipstick urinalysis. Microscopic examination is to be performed only if macroscopic findings are abnormal. Based on Investigator or designee determination or local requirement for participant safety, additional samples may be assessed during the study by a local laboratory, in parallel with the central laboratory.

o For female participants of childbearing potential, a urine hCG test must be performed. Serum pregnancy testing and/or more frequent pregnancy testing will be performed if required by local regulations or in case of suspected pregnancy.

p For post-menopausal female participants (defined as 12 consecutive months of amenorrhea without an alternative medical cause), a serum sampling to test for follicle-stimulating hormone levels must be performed.

q Adverse events will be collected from the time of informed consent through to the last follow-up at the End-of-Study Visit. For participants who complete the 24-week Treatment Period and are entering the open-label Study PHA022121-C307 on the same day, Visit 6 serves as the EOT and EOS Visits.

r Participants will receive a supply of study drug and standard of care on-demand treatment (if needed) at each indicated visit to ensure coverage until the next visit. Supply of study drug and on-demand treatment (if needed) may be provided at the study site, at the participant's home by a visiting nurse or home healthcare provider (for remote visits and if permitted by local regulations), or by direct shipment to the participant's home (where permitted by local regulations).

s Throughout the Treatment Period, it is recommended that participants administer the study drug orally at the same time daily without a meal or with a light meal, with 24 ± 2 hours between each dose when study drug is not administered during a study site visit.
The first study drug administration will occur after all baseline pre-dose assessments are complete at Visit 1. Baseline pre-dose assessments include: completion of ePROs (AE-QoL and AECT-4wk), vital signs, ECGs, clinical laboratory testing, pregnancy testing, PK sampling, adverse events, and of prior and/or concomitant medications. Retraining on the eDiary, ePRO instruments, and other study requirements will be done at study center personnel discretion.
The last scheduled study drug administration for participants who complete the Treatment Period will occur the Day prior to Visit 6 for participants who are entering Study PHA022121-307 and during Visit 6 for participants who are not entering Study PHA022121-C307.
Study drug administration window during on-site or remote visits may be extended to 6 hours (24 ± 6 hours from the last dose) to allow performance of all pre-dose assessments. The participant should use an up to 6-hour window (18 to 24 hours from the last dose) for administration of the study drug on subsequent days to return to their regular dosing schedule.
If a participant misses administration of study drug during their regular dosing time, they should take the study drug as soon as possible on the same Day and should use an up to 6-hour window (18 to 24 hours from the last dose) for administration of the study drug on subsequent days to return to their regular dosing schedule.

t Participants will be required to complete a mandatory eDiary and ePRO training module at the Screening Visit. Retraining on how to use the eDiary will be provided throughout the study, at study center personnel discretion.

u Participants will use an eDiary to record HAE attacks, standard of care on-demand treatments, daily administration of study drug, and to complete ePROs.
Information on HAE attacks will be assessed by the Investigator and/or designee and confirmed by the Investigator within 72 hours from onset of symptoms of an attack. HAE attacks will be recorded from Screening until EOS Visit.
Daily recording of administration of study drug dosing starts from Visit 1 until Visit 6 or EOT Visit (whichever comes first).
ePRO collection will be recorded from Screening until EOT Visit.

v The site staff must instruct participants to complete all ePROs prior to any other assessment performed prior to administration of study drug at the site or during remote or virtual visits.

w To be completed prior to the first dose of study drug as a baseline assessment.

Table 5: Schedule of Pharmacokinetic Sample Collection

Study Participants	Visit 1				Visit 4		Visit 6/EOT [a]
	Pre dose [b]	Post-dose			Pre-dose [b]	Post-dose	Pre-dose [b]
		1 hr ± 15 min	3 hr ± 15 min	5 hr ± 30 min		5 hr ± 30 min	
All adolescents (aged ≥12 to <18 years at the time of signing informed consent)	X	X	X	X	X	X	X
Adults not providing optional consent [c]	X	X					X
Adults providing optional consent [c]	X	X	X	X	X	X	X

Abbreviations: EOT = End of Treatment.

a For participants who complete the 24-week Treatment Period and are entering the open-label Study PHA022121-C307 on the same day, Visit 6 sample collection serves as pre-dose Day 1 of Study PHA022121-C307. For participants who do not complete the 24-week Treatment Period, the Visit 6 sample will be collected at the EOT Visit.

b Pre-dose samples are to be collected within 30 minutes prior to study drug administration.

c The more frequent pharmacokinetic plasma sample collection schedule is optional for adult participants and requires additional consent. Approximately 16 adult participants who provide this consent will undergo the more frequent sampling schedule. All other adult participants will undergo the less frequent sampling schedule.

6.1.2. Study Activities

Study procedures are described below by Visit and in the Schedule of Events (Table 4).

Throughout the study, an eDiary will be completed by participants on a daily basis to record HAE attacks, standard of care on-demand treatments, daily administration of study drug, and to complete ePROs.

6.1.2.1. Screening Period and Run-in Period

Following signing of the ICF, participants will undergo Screening assessments.

The Screening Period will last up to 10 weeks (70 days) to confirm underlying disease status and assess eligibility for study participation. As part of the review of the eligibility criteria, the Investigator will discuss participation in this study and all other available treatment options with the participant. The baseline attack rate will be based on results from the Run-in Period, which occurs during the Screening Period and will last up to 8 weeks (56 days).

The signing of the ICF, Screening Visit, and start of Run-in Period should all occur on the same day.

The following assessments will be performed at the site during the Screening Visit:

- Informed consent
- Training on ePRO instruments (AE-QoL, PGA-Change, AECT-4wk, WPAI-SHP, and TSQM-9) and the eDiary
- Completion of WPAI-SHP and TSQM-9
- Review of eligibility (inclusion/exclusion criteria)
- Demographics and HAE characteristics, including serum sampling for laboratory assessment of C1INH function, C4 level, and C1q level
- Medical history
- Weight and height
- Physical examination
- Vital signs
- Digital triplicate 12-lead ECG
- Clinical laboratory testing (hematology, biochemistry, coagulation, urinalysis)
- Optional biomarkers
- Urine pregnancy test (for women of childbearing potential)
- Follicle-stimulating hormone (FSH) testing (for women who have 12 consecutive months of amenorrhea without an alternative medical cause and are considered post-menopausal)
- Adverse events

- Prior and/or concomitant medications
- Dispensing standard-of-care on-demand medication, if required

The following activities will be performed during the Run-in Period:

- Participant: Daily entry of attack information in the eDiary
- Investigator confirmation of attacks within 72 hours of attack onset
- Weekly contact with participants
  - Days 7, 21, 35, and 49 (± 2 days)
  - Days 14, 28, 42, and 56 (+ 2 days) (ie, on or after the time points for the Run-in Period attack rate eligibility assessment)

As data to confirm eligibility are collected and assessed during the Screening Period, the study site will promptly enter the data as it becomes available (including number of Investigator-confirmed attacks during the Run-in Period) into the electronic case report form (eCRF) for Sponsor review.

6.1.2.2. Day 1 Visit

After confirmation of eligibility, participants will be randomized to receive deucrictibant 40 mg XR tablet or matching placebo QD during the Day 1 Visit. If necessary due to logistical challenges, randomization may take place before the Day 1 Visit; however, no study drug will be dispensed to the participant until Day 1.

Participants cannot be randomized during an HAE attack period. An HAE attack must be resolved for at least 24 hours before randomization.

Other activities performed during the Day 1 Visit prior to administration of study drug include:

- Done first: Completion of ePROs (AE-QoL and AECT-4wk)
- Vital signs
- Digital triplicate 12-lead ECG
- Clinical laboratory testing (hematology, biochemistry, coagulation, urinalysis)
- Urine pregnancy test (for women of childbearing potential)
- PK sampling
- Adverse events
- Prior and/or concomitant medications
- Retraining on the eDiary, ePRO instruments, and other study requirements (eg, AEs, concomitant medications) at study center personnel discretion

Study drug will be provided to the participant after all pre-dose assessments have been completed. The administration of study drug will be documented.

The following will be performed on-site after the participant's first dose of study drug:

- PK sampling
- Standard of care on-demand medication (if required) will be dispensed for the participant to take home.

6.1.2.3. Treatment Period

As noted above, throughout the study, an eDiary will be completed by participants on a daily basis to record HAE attacks, standard of care on-demand treatments, daily administration of study drug, and to complete ePROs.

The sites may resupply participants with study drug and standard of care on-demand medication between scheduled visits.

6.1.2.4. End of Study Activities

- For participants who complete the 24-week Treatment Period and are entering the open-label Study PHA022121-C307 on the same day, Visit 6 serves as the EOT and EOS Visits.
- For participants who complete the 24-week Treatment Period but are not entering the open-label Study PHA022121-C307, Visit 6 serves as the EOT Visit. These participants will proceed to the Follow-up Period and EOS Visit. The EOS Visit occurs 14 to 28 days after Visit 6.
- For participants who do not complete the 24-week Treatment Period, the EOT Visit will occur at the time of study drug discontinuation.
- The Follow-up Period is only for participants who are not entering the open-label Study PHA022121-C307.

6.1.2.5. End of Treatment Visit

For participants who complete the 24-week Treatment Period, Visit 6 will serve as the EOT Visit. For participants who do not complete the 24-week Treatment Period, the EOT Visit will occur at the time of study drug discontinuation.

The following assessments will be performed at the site during the EOT Visit:

- Done first: Completion of ePROs (AE-QoL, PGA-Change, AECT-4wk, WPAI-SHP, and TSQM-9)
- Height, only for participants who were aged ≥12 to <18 years at the time of signing informed consent.
- Weight
- Physical examination
- Vital signs
- Digital triplicate 12-lead ECG
- Clinical laboratory testing (hematology, biochemistry, coagulation, urinalysis)

- Optional biomarkers
- Urine pregnancy test (for women of childbearing potential
- Adverse events
- Prior and/or concomitant medications
- PK sampling

6.1.2.6. End of Study Visit

The EOS Visit assessments will be done for all participants.

The following assessments at the site (or virtual) will be performed at the EOS Visit:

- Adverse events
- Prior and/or concomitant medications

The assessments for the EOT and EOS Visits may be performed at the same visit.

6.2. Rationale for Study Design, Control Group, Study Population, and Dose

This study is designed as a double-blind, placebo-controlled, and randomized study. This design is ethically justified as participants will have adequate access to standard of care on-demand treatment (to be reimbursed or provided by the Sponsor if needed) to treat any HAE attack. Therefore, participants are not placed at any unnecessary risk during this clinical study with adequate clinical monitoring. The placebo-controlled study design had been successfully used in the double-blind placebo-controlled Phase 2 Study PHA022121-C301; there was no increased risk reported to participants. In addition, the placebo-controlled study design has been used and accepted for evaluation of investigational drugs in all ongoing or completed pivotal clinical studies of prophylactic treatment of HAE as of April 2024 (Donidalorsen:Clinicaltrials.gov NCT05139810, 2024; Garadacimab: Craig, 2023; Berotralstat: Zuraw, 2021; Lanadelumab: Banerji, 2018). After completion of the 24-week Treatment Period, participants will be eligible to roll-over into an open-label Study PHA022121-C307, to receive active deucrictibant prophylactic treatment for long-term safety and efficacy assessment.

In addition to adult participants enrolled in Phase 2 Study PHA022121-C301, adolescent participants (≥12 to <18 years of age) with confirmed diagnosis were also eligible to enroll. The inclusion of adolescents is justified by the lack of any significant safety or tolerability issues identified in the completed nonclinical and clinical studies with deucrictibant, robust efficacy findings of significant reduction of HAE attack rate demonstrated in the Phase 2 prophylactic PHA022121-C301 study, and PK analysis suggesting that deucrictibant exposure in participants with HAE is comparable to what has been observed in healthy participants. Previous population PK has shown that age and weight are not significant covariates of deucrictibant PK. In addition, there have been no known differences in disease presentation and pathophysiology reported between adults and adolescents with HAE. Therefore, it is projected that deucrictibant exposure level in adolescents will be within the same range as for adult participants. The inclusion of adolescents in PHA022121-C305 study can be well justified by the favorable benefit to risk assessment.

In this study, deucrictibant 40 mg XR tablets QD will be evaluated. In the Phase 1 Study PHA022121-C015, the median pre-dose deucrictibant plasma levels at steady state (Day 5) of deucrictibant 40 mg XR tablets QD was 43.4 ng/mL, which is approximately twice the steady state pre-dose levels obtained with deucrictibant 20 mg soft capsules BID in Study PHA022121-C301; meanwhile, the median maximal deucrictibant plasma level at steady state (Day 5) of deucrictibant 40 mg XR tablets QD was 106 ng/mL, which is approximately 68% of the steady state maximal level obtained with deucrictibant 20 mg soft capsules BID in Study PHA022121-C301. Therefore, compared to the deucrictibant 20 mg soft capsules BID, deucrictibant 40 mg XR tablets QD can be reasonably expected to provide comparable or better benefit-risk profiles. In addition, once daily administration is also more convenient for patients and can lead to better treatment adherence.

6.3. Appropriateness of the Measurements

All efficacy, safety, and PK measurements in the study are widely used and generally recognized as reliable, accurate, and relevant.

6.4. Study Duration and Dates

The Screening Period will last up to 10 weeks in duration. The baseline attack rate will be determined based on the results of the Run-in Period, which occurs during the Screening Period and will last up to 8 weeks. The Treatment Period is 24 weeks in duration and the Follow-up Period is up to 4 weeks in duration; the Follow-up Period may not occur if the participant enrolls into the open-label study PHA022121-C307. Therefore, the duration of an individual's participation in this study will be no longer than 38 weeks.

6.5. Number of Participants

Approximately 81 participants (male or female, age at Screening ≥12 years), including at least 8 adolescents (male or female, age at Screening ≥12 to <18 years), with HAE will be randomized; see Section 13.2.

Among the approximately 81 participants, approximately 75 participants with HAE type 1 or 2 and approximately 6 participants with HAE type 3 are expected to be randomized.

6.6. Treatment Assignment

See Section 8.5.

6.7. Dose Adjustment and Interruption Criteria

This is a daily single-dose study and no changes to dose are permitted.

Treatment can be temporarily interrupted for safety concerns or other reasons at the Investigator's discretion and in consultation with the Medical Monitor.

6.8. Criteria for Study Termination

This study will be prematurely terminated or suspended by the Sponsor if there is sufficient reasonable cause. This may include reasons related to the benefit/risk assessment of participating

in the study, practical reasons (including slow enrollment or business considerations), or for regulatory or medical reasons. In taking the decision to terminate, the Sponsor will consider the participant's welfare and safety. Circumstances that warrant termination or suspension include, but are not limited to:

- Determination of unexpected, significant, or unacceptable risk to participants (eg, new and unjustifiable risk and/or toxicity in risk-benefit analysis)
- Serious or severe neurological adverse events. If serious or severe neurological events with same Preferred Term (PT) or High Level Term regardless of relatedness to study drug are reported by 2 or more participants in the same study, the Independent Data Monitoring Committee (IDMC), including a neurologist, will be convened immediately by the Sponsor to further evaluate the risk and causality and decide if there is sufficient reasonable cause to recommend early termination of the study.

Should early termination of the study be necessary, participants must be seen as soon as possible (instructions will be provided to the Investigator for contacting the participant, when the participant should stop taking study drug, and when the participant should come for a final visit) and treated as a prematurely withdrawn participant. The Investigator may be informed of additional procedures to be followed in order to ensure that adequate consideration is given to the protection of the participant's interests. The Investigator or Sponsor, depending on the local regulations, will be responsible for informing Institutional Review Boards (IRBs)/Independent Ethics Committees (IECs) of the early termination of the study.

6.9. Study Completion

The study will be considered complete when all participants have completed the study, have withdrawn from the study, or have enrolled into the open-label study PHA022121-C307 (see Section 6.1).

7. SELECTION AND WITHDRAWAL OF PARTICIPANTS

The study is planned to be conducted at approximately 80 study sites in approximately 30 countries.

7.1. Participant Inclusion Criteria

Participants must meet all of the following criteria to be eligible to participate in this study:

1. Provision of written informed consent. At the time of signing informed consent, male and female participants must be aged ≥12 years. Adolescent participants (ie, aged ≥12 to <18 years or as determined by local law) must also be ≥40 kg in body weight at Screening. If the participant is an adolescent, written assent will be obtained from the participant and consent will be obtained from the participant's parent/legally designated representative/guardian, per local regulations. A participant who is an adolescent will sign the ICF if they reach the age of 18 years or as determined by local law during their participation in the study.

2. Participants who previously completed Study PHA022121-C306 may be eligible to be screened for this study. For these participants, there is no need to repeat HAE diagnostic test as described in Inclusion Criterion #3 if the C1INH function was confirmed by the central laboratory in Study PHA022121-C306.
3. Diagnosis of HAE type 1/2 or type 3 based on the following:
   a. For participants with HAE type 1/2:
      - Documented clinical history consistent with HAE (cutaneous or submucosal, nonpruritic swelling without accompanying urticaria)
      - At least one of the following:
        – Age at reported onset of first angioedema symptoms ≤30 years
        – Family history consistent with HAE
        – C1q above the lower limit of the normal range by central laboratory as part of the Screening assessments
      - Diagnostic testing results to confirm HAE:
        – C1INH functional level <50% of the normal level must be shown by chromogenic assay performed by the central laboratory as part of the Screening procedures. Participants with functional C1INH level 40% to <50% of the normal level must also have a C4 level below the normal range.
   b. For participants with HAE type 3:
      - Recurrent angioedema attacks with diagnostic testing results obtained during Screening to confirm C1INH function ≥50% of normal and C4 level not below the lower level of the normal range performed by the central laboratory
      - Must meet one of the following:
        – Documented genetic mutation associated with HAE type 3 as listed in the HAEA and WAO/EAACI Guidelines (Appendix 2)

          Or

        – If no documented genetic mutation:
          - Clinical diagnosis with family history of HAE type 3 and an elevated BK peptide level previously confirmed by a commercially available liquid chromatography-tandem mass spectrometry (LC-MS/MS) assay

            And

          - Attacks not responding to treatments with high-dose antihistamine (cetirizine 40 mg/day or equivalent high-dose second-generation antihistamine medication) and no clinical attack symptom relief if treated with corticosteroid, montelukast, or omalizumab
      - Documented effective attack symptom relief with on-demand icatibant treatment

Note: The C1INH function sample should be collected at least 5 half-lives after the last dose of C1INH therapy, if applicable.

4. History of at least 3 HAE attacks within the 3 consecutive months prior to Screening Visit.

   Note: For participants taking long-term prophylactic HAE therapy before Screening, at-least 3 attacks may be within the last 3 consecutive months before commencing the prophylactic treatment.

5. Participants must experience [redacted] Investigator-confirmed attacks during the up to 8-week Run-in Period. Randomization should take place after any of the following baseline attack criteria has been met:

   - [redacted]

   Note: The baseline attack rate requirement for eligibility should not be disclosed to study participants.

6. Participant is assessed by the Investigator to have reliable access and ability to use standard of care on-demand treatments to effectively manage acute HAE attacks.
7. Investigator considers that the participant (and parent/legally designated representative/guardian for adolescent participants) is willing and able to adhere to all protocol requirements, including ensuring that the participant is capable of, and compliant with, eDiary and ePRO data recording.
8. Female participants of childbearing potential (or who become of childbearing potential during the study) must agree to the protocol specified pregnancy testing and to be abstinent from heterosexual intercourse or to use a highly effective contraception method as defined in the protocol (Section 8.2.2) and as available locally, from enrollment until 30 days after the last study drug administration.

   There are no contraceptive requirements for male participants.

7.2. Participant Exclusion Criteria

Participants who meet any of the following criteria will be excluded from the study.

1. Any diagnosis of angioedema other than HAE
2. Participation in a clinical study with any other investigational drug within the last 30 days or within 5 half-lives of the investigational drug at Screening (whichever was longer)

3. Has received prior prophylactic treatment or on-demand treatment with deucrictibant, with the exception of participants from Study PHA022121-C306
4. Exposure to angiotensin-converting enzyme (ACE) inhibitors or any estrogen-containing medications with systemic absorption (such as oral contraceptives or hormonal replacement therapy) within 4 weeks of Screening
5. Prior gene therapy for any indication at any time
6. Receiving prophylactic treatment for HAE and are satisfied with this treatment. Patients who are not satisfied (eg, tolerability issues, lack of efficacy) and have previously stopped long-term prophylactic HAE treatment can sign the ICF and begin the Screening Period only if their last dose of treatment was received prior to the time point before Screening indicated below:
    a. Long-term prophylactic therapy for HAE with C1INH, oral kallikrein inhibitors, or anti-fibrinolytics: 2 weeks prior to Screening
    b. Long-term prophylactic therapy for HAE with attenuated androgens: 4 weeks prior to Screening
    c. Long-term prophylactic monoclonal antibody therapy for HAE (ie, lanadelumab): 5 half-lives prior to Screening
    d. Short-term prophylaxis for HAE: 7 days prior to Screening. Note: Short-term prophylaxis is defined as iv C1INH to avoid angioedema complications from medically indicated procedures.

    Note: If prophylaxis is not listed above, please consult with Sponsor.
7. Any females who are pregnant, plan to become pregnant, or are currently breast-feeding
8. Abnormal hepatic function (aspartate aminotransferase [AST] >2× upper limit of normal [ULN], alanine aminotransferase [ALT] >2× ULN, or total bilirubin >1.5× ULN or any hepatic impairment via Child-Pugh Scoring System). Participants with Gilbert's syndrome, defined as isolated increase of total bilirubin ≤3× ULN and AST and ALT within the normal range, are not excluded.
9. Moderate or severe renal impairment (estimated glomerular filtration rate [eGFR] <60 mL/min/1.73 m^2)
10. Any clinically significant history of angina, myocardial infarction, syncope, stroke, left ventricular hypertrophy or cardiomyopathy, uncontrolled hypertension, bradycardia, or any other clinically significant cardiovascular abnormality within the previous year that, in the opinion of the Investigator, would interfere with the participant's safety or ability to participate in the study
11. History of epilepsy and/or other significant neurological diseases
12. Any clinically significant and uncontrolled gastrointestinal dysfunction (eg, chronic diarrhea, inflammatory bowel disease) which impacts study drug absorption
13. History of alcohol or drug abuse within the previous year, or current evidence of substance dependence or abuse

14. Use of concomitant medications with systemic absorption and foods that are moderate and strong inhibitors of cytochrome P450 (CYP)3A4 such as clarithromycin, erythromycin, diltiazem, itraconazole, ketoconazole, ritonavir, verapamil, and grapefruit juice or strong inducers of CYP3A4 such as carbamazepine, phenytoin, and rifampin within the last 30 days or within 5 half-lives (whichever is longer) of the time of randomization
15. Known hypersensitivity to deucrictibant or any of the excipients of the study drug

7.3. Discontinuation of Study Treatment and Participant Withdrawal

The reason for participant discontinuation of study treatment or withdrawal from the study will be recorded in the eCRF and documented in the participant's medical record.

Participants may discontinue study drug or withdraw from the study at any time at their own request, or the Investigator and/or Sponsor may discontinue a participant from the administration of study drug or from the study for any of the following reasons:

- Pregnancy
- Significant study drug or procedure noncompliance
- Lack of effectiveness
- Adverse events: If any clinical AE/SAE, laboratory abnormality, serious or severe neurological AE, or other medical condition or situation occurs that negatively impacts the benefit-risk in such a way that continued administration of study drug would not be in the best interest of the participant. If such an event is reported, study treatment will be stopped immediately, and causality of the event will be evaluated. The study drug treatment will be discontinued permanently if the event is suspected to be related to study drug.
- Sponsor termination of the study
- Investigator considers it is in the participant's best interest to withdraw
- Participant request

Participants who discontinue study drug before Week 24 of the Treatment Period will complete the End of Treatment (EOT) Visit and will be encouraged to remain in the study per study protocol defined visit schedule through the End of Study (EOS) Visit.

Participants who withdraw early from the study will be encouraged to complete the EOT and EOS Visits after their last administration of study drug. The assessments for both visits may be done at the same time. For participants who have already discontinued study drug and completed the EOT Visit at an earlier time, only the EOS Visit needs to be completed when they withdraw from the study.

If the participant withdraws from the study, and also withdraws consent for disclosure of future information, the Sponsor may retain and continue to use any data and samples collected before withdrawal of consent.

7.3.1. Lost to Follow-up

If a participant repeatedly fails to return for scheduled visits, the site must attempt to contact the participant and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether or not the participant wishes to and/or should continue in the study. All attempts to contact the participant should be documented in the participant's medical record. A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and is unable to be contacted by the study site. Participants lost to follow-up will be considered to have withdrawn from the study.

7.3.2. Replacement Policy

Participants will not be replaced in this study.

8. TREATMENT OF PARTICIPANTS

8.1. Description of Study Drug

A description of the deucrictibant 40 mg XR tablet and placebo tablet is provided in Section 9.1.1 and Section 9.1.2, respectively.

8.2. Prior and Concomitant Medications

Prior non-HAE therapies and HAE treatments used by the participant within 30 and 90 days, respectively, before the Screening Visit must be documented as such in the eCRF.

All concomitant HAE and non-HAE treatments (including but not limited to all prescription, over-the-counter medications, vitamins and supplements, herbal treatments, non-drug supportive therapies and procedures) administered to participant from the time of signing the ICF through the final study contact (ie, End of Study Visit) are considered concomitant therapies and must be documented as such in the eCRF, including the start and stop dates, dose, frequency, and indication.

8.2.1. Management of HAE Attacks

Participants must have access to standard of care on-demand treatments and be able to treat any HAE attack that occurs. See Section 10.1 and Section 10.2 for information regarding assessing and reporting HAE attacks.

Participants should continue administering study drug and following study procedures without alteration to the protocol schedule, even if they receive on-demand treatment for HAE attacks. The use of standard of care on-demand medications associated with the management of HAE attacks must be recorded in the eDiary.

8.2.2. Contraception

Following completion of the EFD studies in pregnant animals, deucrictibant is considered a study drug with unlikely human teratogenicity/fetotoxicity in early pregnancy according to the Clinical Trials Coordination Group recommendations (CTCG, 2024).

Female participants of childbearing potential require the use of medically accepted, highly effective contraceptive measures from enrollment until 30 days after the last study drug administration.

The following birth control methods are considered highly effective:

- Progestogen-only hormonal contraception associated with inhibition of ovulation: oral, injectable, or implantable
- Intrauterine device (IUD, all types)
- Intrauterine hormone-releasing system (IUS)
- Bilateral tubal occlusion
- Sexual abstinence, where it is the anticipated usual lifestyle of the participant
- Vasectomized partner (provided that partner is the sole sexual partner and that the vasectomized partner has received medical assessment of the surgical success.)

Females of non-childbearing potential, defined as prepubertal, surgically sterile (status post hysterectomy, bilateral oophorectomy, or bilateral salpingectomy), or post-menopausal (defined as no menses for at least 12 months without an alternative medical cause and an FSH test result indicative of post-menopausal status) do not require contraception during the study.

There are no contraceptive requirements for male participants.

Note: Not all methods acceptable for use during participation in this study are approved by local regulations. The selection of a contraceptive method should be based on the approved methods in the participant's country.

8.2.3. Prohibited Concomitant Medications

The following concomitant therapies are prohibited during study participation:

- Any medications for long-term prophylactic use to prevent HAE attacks including kallikrein inhibitors, antifibrinolytics (tranexamic acid or epsilon-aminocaproic acid), attenuated androgens (eg, danazol, stanozolol, and oxandrolone), or C1INH concentrates. If a prophylactic therapy is not listed above, please consult with Sponsor.

  Note: Short-term prophylactic HAE treatment is permitted during the Treatment Period. Short-term prophylaxis is defined as iv C1INH to avoid angioedema complications from medically indicated procedures.

- Use of concomitant medications with systemic absorption and foods that are moderate or strong inhibitors of CYP3A4, such as clarithromycin, erythromycin, diltiazem, itraconazole, ketoconazole, ritonavir, verapamil, and grapefruit or strong inducers of CYP3A4, such as carbamazepine, phenytoin, and rifampin within the last 30 days or within 5 half-lives (whichever is longer) at the time of enrollment. If a participant requires a short-term treatment with a moderate or strong inhibitor or strong inducer of CYP3A4 for an intercurrent illness or a procedure, the Medical Monitor should be consulted to advise on whether treatment with study drug should be suspended and a washout period is needed and, if it is, for how long.

- Angiotensin converting enzyme inhibitors
- Estrogen-containing medications with systemic absorption
- Any other investigational drug or device

The use of medications that are substrates of P-gp and BCRP transporters is not prohibited.

8.3. Restrictions

8.3.1. Fluid and Food Intake

No restrictions for fluid and food intake apply during the study other than moderate or strong inhibitors of CYP3A4 (eg, grapefruit).

8.3.2. Participant Activity Restrictions

No restrictions on participant activity apply during the study.

8.4. Treatment Compliance

Participants will record daily in the eDiary the administration of the study drug.

In addition, participants will be instructed to bring unused study drug and empty/used study drug packaging to every on-site visit. Treatment compliance will be assessed by the study personnel documenting details on used and unused study drug.

8.5. Randomization and Blinding

Participants cannot be randomized during an HAE attack period. An HAE attack must be resolved for at least 24 hours before randomization.

Eligible participants will be randomized in a 2:1 ratio to receive deucrictibant 40 mg XR tablet or matching placebo QD for 24 weeks. Stratification factors of the randomization will be age group (adolescents [≥12 to <18 years] and adults [≥18 years]) and baseline HAE attack rate [redacted] per 4 weeks versus [redacted] attacks per 4 weeks).

Participants, Investigators and site personnel, and the Sponsor will be blinded to treatment assignment.

Blinding of the treatment administered is achieved with the identical appearance of deucrictibant tablets and placebo tablets. Each administered dose consists of 1 tablet of deucrictibant 40 mg XR or matching placebo tablet as described in Section 6.1.

The study labels are blinded and contain a unique code. Assignment of study drug in accordance with randomized treatment is managed by the interactive response technology (IRT) system (see Section 9.5).

Access to treatment assignment will be available if the Investigator or designee deems it necessary to break the study blind in the interest of a participant's medical safety, in case of a medical emergency, to meet regulatory reporting obligations, or if warranted during scheduled safety reviews. The Investigator will have the option of contacting the Medical Monitor to discuss the situation which has arisen and resulted in the need for unblinding of the participant.

However, unblinding procedures can be initiated by the Investigator without the need to contact the Medical Monitor in an emergency situation. The Medical Monitor will not be involved in the decision to unblind.

The IDMC, in consultation with the Medical Monitor, may also request unblinding of an individual participant's treatment assignment for safety reasons.

In the event of a drug-related, serious, unexpected AE, the Sponsor's Drug Safety department/designee will be provided with the participant's treatment assignment for the purpose of regulatory reporting. The treatment code will be broken before reporting each expedited event (eg, suspected unexpected serious adverse reaction [SUSAR]) to regulatory authorities. The IRT will be used by the Drug Safety department/designee to perform regulatory unblinding to identify treatment allocation for each potentially expedited report (ie, those that meet the criteria for unblinding, which includes SAEs that are related and unexpected). The treatment allocation will be filed in a secure electronic repository.

Sites will be informed of each participant's treatment allocation once the study is completed.

9. STUDY DRUG MATERIALS AND MANAGEMENT

9.1. Study Drug and Standard of Care On-Demand Medication

Study drug refers to both deucrictibant and the matching placebo tablet, administered as blinded study drug.

9.1.1. Active Study Drug

Active study drug is a tablet for oral administration containing 40 mg deucrictibant.

A description of deucrictibant 40 mg XR tablet is presented in Table 6.

Table 6: Description of Deucrictibant 40 mg Extended-release Tablet

Product Name	Deucrictibant extended-release (XR) tablet
Dosage Form	Film-coated yellow, oval, biconvex shaped tablet
Unit Dose	Each tablet contains 40 mg of deucrictibant
Route of Administration	Oral
Physical Description	Film-coated yellow, oval, biconvex shaped tablet with debossing "PHA" on one side

9.1.2. Placebo

To maintain the study blind during the Treatment Phase, matching placebo for oral administration will be used.

9.1.3. Standard of Care On-Demand Medication

If HAE attacks occur during the study, participants should use standard of care on-demand medication that is approved in their country for the on-demand treatment of HAE attacks. The Sponsor may reimburse the use of standard of care on-demand medication.

Alternatively, commercially available icatibant may be provided in exceptional cases. The commercially available icatibant will be labelled according to the applicable regulations for the country in which it is used.

9.2. Study Drug Packaging and Labeling

The study drug (deucrictibant or placebo) is packaged according to current Good Manufacturing Practice.

The study drug (deucrictibant or placebo) is packaged in aluminium/aluminium blisters. Blisters are sealed into a wallet and the wallet is labelled and rendered tamper evident.

9.3. Study Drug Storage

The Investigator or designee has overall responsibility for ensuring that the study drug is stored in a secure, limited-access location. Access to the study drug should be restricted to designated study personnel.

Participants will be instructed on appropriate storage of the study drug at their home location.

The study drug should not be stored above 25ºC (do not freeze). Further details on study drug storage will be provided in the pharmacy instructions provided separately.

9.4. Study Drug Administration

The QD administration of study drug throughout the Treatment Period should occur at the same time daily without a meal or with a light meal, with 24 ± 2 hours between each dose when study drug is not administered during a study visit. The tablets are to be swallowed with water.

Study drug will be provided to the participant after all baseline pre-dose assessments have been completed at Visit 1. The administration of study drug will be documented.

The last scheduled study drug administration for participants who complete the Treatment Period will occur the Day prior to Visit 6 for participants who are entering Study PHA022121-C307 on the same Day as Visit 6, and during Visit 6 for participants who are not entering Study PHA022121-C307 or who are entering Study PHA022121-C307 at a later time.

Study drug administration window during on-site or remote visits may be extended to 6 hours (24 ± 6 hours from the last dose) to allow performance of all pre-dose assessments. The participant should use an up to 6-hour window (18 to 24 hours from the last dose) for administration of the study drug on subsequent days to return to their regular dosing schedule.

If a participant misses administration of study drug during their regular dosing time, they should take the study drug as soon as possible on the same Day and should use an up to 6-hour window (18 to 24 hours from the last dose) for administration of the study drug on subsequent days to return to their regular dosing schedule.

9.5. Study Drug Dispensing and Accountability

Participants will receive a supply of study drug and on-demand treatment (if needed) at each indicated visit to ensure coverage until the next visit. Supply of study drug and on-demand treatment (if needed) may be provided at the study site, at the participant's home by a visiting nurse or home healthcare provider (for remote visits and if permitted by local regulations), or by direct shipment to the participant's home (where permitted by local regulations).

The study drug must be dispensed under the supervision of the Investigator or designee using the IRT system. The Investigator or designated pharmacist will maintain up-to-date accountability records recording receipt, dispensing, return, and destruction (Section 9.6) of all clinical study supplies. Local regulations must be adhered to.

Any damaged or unusable study drug in a given shipment will be documented in the study files and recorded in IRT. The Investigator must notify the Sponsor of any damaged or unusable study drug that was supplied to the Investigator's site.

Any medication errors, including incorrect dosage, use of expired study drug, storage temperature excursions, administration errors, etc should be recorded as protocol deviations and the Sponsor should be informed immediately.

9.6. Study Drug Handling and Disposal

All unused study drug and empty/used study drug packaging will either be returned to the Sponsor (or nominated facility) or can be destroyed on site (according to local regulations) or by an authorized destruction unit following written agreement with the Sponsor. All correspondence related to study drug destruction must be filed with the study accountability records.

10. ASSESSMENT OF EFFICACY

Analyses of the efficacy endpoints in this study require documentation of specific information regarding the participant's symptoms of any HAE attacks experienced during the study.

10.1. Definition of an HAE Attack

The criteria provided in this section should be used by the Investigator to assess if an event reported by a participant can be confirmed as an HAE attack.

To be confirmed as an HAE attack, an event must have signs or symptoms consistent with an attack in at least 1 of the following locations:

- Peripheral angioedema: cutaneous swelling involving an extremity, face, neck, torso and/or genitourinary region
- Abdominal angioedema: abdominal pain, with or without abdominal distension, nausea, vomiting, or diarrhea
- Upper airway angioedema: sensation of lump in the throat, difficulty swallowing, throat tightening, difficulty speaking, hoarseness, stridor, dyspnea, or upper airways swelling on the tongue, palate, uvula, or larynx

Despite the presence of these symptoms, the Investigator may still determine that the event does not represent an HAE attack if there are clinical features that strongly refute such a diagnosis (eg, associated with wheals or longer in duration than the typical time course of an attack or alternate etiology such as recent trauma resulting in swelling or pain).

To be counted as a unique HAE attack distinct from a previous attack, the new symptoms must occur at least 24 hours after complete resolution of symptoms of a prior attack (ie, there must be a 24-hour symptom-free period to be considered a new attack).

Participant-reported occurrences of prodromal symptoms alone or pre-emptive use of acute treatments without swelling symptoms are not considered to meet the criteria of an HAE attack.

10.2. Reporting of HAE Attacks

Efficacy will be assessed based upon the occurrences of HAE attacks, which will be confirmed by Investigators. During the Screening Visit, site personnel will discuss with participants the requirements for reporting HAE attacks, including the information that is to be recorded in the eDiary for each HAE attack.

An electronic Clinical Outcomes Assessment (eCOA) solution for use on an electronic device or application will be used for entry of the patient-reported outcome data by the participant into a study-specific eDiary. Provisions will be made for back-up method(s) in case of device malfunction. The ePRO data and back-up data will be considered source data.

Participants will receive training on how to use the eDiary at the Screening Visit and throughout the study, as needed. Information on HAE attacks will be collected in the eDiary by the participant until the completion of the study. Each reported HAE attack will be assessed by the Investigator and/or designee and confirmed by the Investigator within 72 hours from onset of symptoms of an attack.

The HAE attack data outlined in this section and the criteria in Section 10.1 should be used by the Investigator to assess if an event reported by a participant can be confirmed as an HAE attack.

The following information will be entered by the participant in the eDiary:

- Date and time of onset of symptoms of an HAE attack
- Date and time of complete resolution of attack symptoms
- If standard of care on-demand treatment was administered for the HAE attack

Before the Investigator and/or designee completes a separate Investigator Assessment Form for each participant-reported HAE attack, they will review the data entered by the participant via the eDiary vendor portal and follow-up with the participant. Follow-up topics include, but are not limited to:

- Location of attack symptoms
- Start/end date/time of symptoms
- Attack severity

  - Mild: the HAE attack is noticeable but is easily tolerated by the participant and does not interfere with routine activities
  - Moderate: the HAE attack limits/interferes with the participant's ability to attend work/school or participate in family life and social/recreational activities
  - Severe: the HAE attack significantly limits the participant's ability to attend work/school or participate in family life and social/recreational activities
- If standard of care on-demand medications were administered for the HAE attack (date, time, and medication(s) including the number of doses administered)
- Type of medical assistance or intervention, if required
- If hospitalization or a visit to emergency department occurred

Additional queries by the Investigator and/or designee during participant follow-up may include questions such as: whether the participant only experienced prodromal symptoms, whether the symptoms experienced were typical or different from prior attacks, or whether there were possible alternative etiologies for the symptoms.

The outcome of the follow-up will determine if an HAE attack is confirmed. Any participant-reported HAE attack that is not confirmed by the Investigator as an HAE attack must have an alternate AE diagnosis recorded.

HAE attacks and their associated symptoms will not be defined as AEs/SAEs per se. Some of the triggers for HAE attacks, such as an infection or trauma, may be considered as AEs or SAEs, and therefore should be reported per the usual AE reporting process. HAE attacks that meet criteria as defined in Section 12.9.1.2 are considered SAEs and should be reported per the usual SAE reporting process.

11. Patient Reported Outcomes

The ePRO assessments are to be completed within the eDiary at the time points indicated in the Schedule of Events (Table 4). The site staff must instruct participants to complete all ePROs prior to any other assessment performed prior to administration of study drug at the site or during remote or virtual visits.

11.1. Angioedema Quality of Life Questionnaire

The Angioedema Quality of Life (AE-QoL) questionnaire is a symptom-specific tool developed and validated to assess health-related quality of life impairment in patients with recurrent angioedema (Weller, 2012). The AE-QoL consists of 17 questions across 4 domains/dimensions (functioning, fatigue/mood, fear/shame, nutrition). Each of the 17 items has a 5-point Likert-type response scale ranging from 1 (Never) to 5 (Very Often). The AE-QoL is scored to produce a score for each domain and a total score ranging from 0 to 100, with higher scores indicating greater impairment.

11.2. Patient Global Assessment of Change

One anchor question will be completed by participants, the Patient Global Assessment of Change (PGA-change). The PGA-change item covers the overall change in how the patient's QoL has been impacted by their HAE since starting to take the study drug with a 5-point Likert response scale (much better, a little better, no change, a little worse, or much worse). It will be used to support the interpretation of other patient reported outcomes.

11.3. Angioedema Control Test

The Angioedema Control Test (AECT) was developed and validated to retrospectively quantify disease control and to aid treatment decisions in patients with recurrent angioedema (Weller, 2020a, Weller, 2020b). The AECT has 4 questions with each question having 5 answer options (each scored 0 to 4 points), addressing the frequency of angioedema, angioedema related QoL impairment, unpredictability of angioedema attacks, and angioedema control by the current treatment approach. The AECT ranges from 0 to 16 points, with total scores of 0 to 9 points indicating poorly controlled disease and 10 to16 points indicating well controlled disease. The AECT version with a 4-week recall period (ie, AECT-4wk) will be used in this study.

11.4. Treatment Satisfaction Questionnaire for Medication

Developed in 2007, the Treatment Satisfaction Questionnaire for Medication 9-items (TSQM-9) is a streamlined version of the longer TSQM 1.4 and evaluates patient treatment satisfaction with the medication for the following scales: effectiveness, convenience, and overall satisfaction. Scale scores are transformed into scores ranging from 0 to 100 and can be used to calculate a total composite score, with higher scores indicating greater satisfaction (Bharmal, 2009).

11.5. Work Productivity and Activity Impairment Questionnaire: Specific Health Problem

The Work Productivity and Activity Impairment Questionnaire: Specific Health Problem (WPAI-SHP) is a tool used to assess how a specific health condition impacts a person's ability to work and do regular activities. It measures absenteeism, presenteeism, and impairments in unpaid activity due to health problems during the past seven days (Reilly, 1993).

12. SAFETY AND PHARMACOKINETICS ASSESSMENTS

Safety and PK assessments are to be completed at the time points indicated in the Schedule of Events (Table 4 and Table 5).

12.1. Informed Consent

At the Screening Visit, a signed ICF, and assent form, if applicable, must be obtained from each participant or participant's parent/legally designated representative/guardian before performing any study-related procedures. Each participant will be given both verbal and written information describing the nature and duration of the clinical study. Further details on participant consent are provided in Section 16.2.

If an adolescent participant reaches the age of 18 years or as determined by local law during study participation, the participant will be required to provide written informed consent in order to continue study participation, in accordance with local regulations.

12.2. Demographics and HAE Characteristics

At Screening, demographic information, including year of birth, gender, and, where permitted, race and ethnicity, will be captured in the eCRF for each participant. Demographic data may be used to facilitate subgroup analyses of study data.

In addition, HAE characteristics, including the year of diagnosis, family history, the type of HAE, the number of attacks in the 3 consecutive months prior to Screening Visit or in the 3 consecutive months prior to previous long-term prophylactic medication, as well as the number of attacks during the Run-in Period, and the type of non-localized symptoms typically associated with the participant's HAE attacks will be collected.

12.3. Medical History

At Screening, existing relevant medical history (current or resolved at the time of signing the ICF) will be recorded for the following body systems: head/eyes/ears/nose/throat, respiratory, cardiovascular, gastrointestinal, genitourinary, musculoskeletal, neurological, endocrine/metabolic, blood/lymphatic, dermatologic, psychiatric, allergy/immunology, and other.

12.4. Weight and Height

Height will be measured at the Screening Visit for all participants. At Visit 6 (or the EOT Visit, if the participant discontinues study drug early), height will also be measured for participants who were aged ≥12 to <18 years at the time of signing informed consent.

Weight will be measured at the Screening Visit and at other time points specified in the Schedule of Events.

12.5. Physical Examination

A physical examination will be conducted at the Screening Visit and at other time points specified in the Schedule of Events. The following body systems will be examined: gastrointestinal system; head/eyes/ears/nose/throat; endocrine (including thyroid); respiratory system; cardiovascular system; integumentary system (skin); musculoskeletal; and nervous system (central and peripheral). Additional physical examinations may be scheduled at the discretion of the Investigator (or designee).

If any abnormalities in these systems are identified, further examinations or external consultations are required. Clinically significant findings should also be documented on the eCRF.

12.6. Vital Signs

Vital signs include blood pressure, heart rate, and body temperature. Vital signs will be measured at the Screening Visit and at other time points specified in the Schedule of Events. Blood

pressure (systolic and diastolic) and heart rate will be taken after the participant has rested for at least 5 minutes. Blood pressure measurements must be obtained with an appropriate cuff size and with the participant's arm supported at the level of the heart. It is acceptable to obtain a heart rate from the blood pressure instrument or the ECG machine.

12.7. Electrocardiograms

A digital 12-lead ECG (recorded in triplicate) will be performed and recorded after the participant has rested in the supine position for at least 5 minutes at the Screening Visit and at other time points specified in the Schedule of Events. Central reading of all recorded ECGs will be performed by the ECG vendor. ECGs will be evaluated by the Investigator or designee based on the central reading. If the reading is abnormal, the Investigator will assess whether it is "Clinically significant" or "Not clinically significant" and record the assessment in the eCRF.

ECG recordings should occur prior to the PK sample collection, if applicable.

12.8. Clinical Laboratory Tests

12.8.1. Safety Laboratory Parameters

Safety laboratory testing will be performed at Screening and at other time points specified in the Schedule of Events. Clinical laboratory tests for hematology, biochemistry, coagulation, and urinalysis are indicated in Table 7. Females of childbearing potential must have urine pregnancy testing performed at the Screening Visit and at other time points specified in the Schedule of Events. Serum pregnancy and/or more frequent pregnancy testing will be performed if required by local regulations or in case of suspected pregnancy.

Unscheduled clinical laboratory testing is permitted if deemed necessary for safety by the Investigator or designee and/or Sponsor. Additional safety sampling may be performed and processed by a local laboratory at the same time points as the central laboratory.

Participants will be in a seated or supine position during blood collection. Blood and urine samples will be processed by a central laboratory, with the exception of urine pregnancy testing. The Investigator will classify all laboratory values outside the normal range of the central laboratory as "Clinically significant" or "Not clinically significant".

Table 7: List of Safety Laboratory Tests

Hematology: • Hematocrit (Hct) • Hemoglobin (Hgb) • Mean corpuscular hemoglobin (MCH) • Mean corpuscular hemoglobin concentration (MCHC) • Mean corpuscular volume (MCV) • Platelet count • Red blood cell (RBC) count • White blood cell (WBC) count with differential count (absolute and relative values) Coagulation: • Prothrombin time • International normalized ratio (INR) Urinalysis: • Macroscopic analysis: • Bilirubin • Blood • Glucose • Ketones • Leukocyte esterase • Nitrite • pH • Protein • Specific gravity • Urobilinogen • Microscopic analysis (reflex only): • Bacteria • Casts • Crystals • Epithelial cells • RBCs • WBCs • Yeast	Biochemistry: • Albumin • Alkaline phosphatase (AP) • Alanine aminotransferase (ALT; SGPT) • Aspartate aminotransferase (AST; SGOT) • Blood urea nitrogen (BUN) • Bicarbonate • Calcium • Chloride • Creatinine • Gamma-glutamyl transferase (GGT) • Glucose • Lactate dehydrogenase (LDH) • Magnesium • Phosphate • Potassium • Sodium • Total bilirubin • Direct bilirubin • Total protein • Uric acid • eGFR (CKD-EPI ≥18 years of age or modified Schwartz 12 to <18 years of age, calculated) Pregnancy test (for women of childbearing potential): • Urine human chorionic gonadotropin (hCG) Follicle-stimulating hormone (FSH) test (for confirming post-menopausal status) • Serum FSH

12.8.2. Pharmacokinetic Sample Collection, Storage, and Shipping

In all participants, PK plasma samples will be collected at Visit 1 pre-dose (within 30 min prior to the dose) and at 1 hour (± 15 min) post-dose and at Visit 6 pre-dose (within 30 min prior to the dose).

In approximately 16 adult participants who consent to additional sampling and all adolescent participants, PK plasma samples will also be collected at 3 hours (± 15 min), and 5 hours (± 30 min) post-dose at Visit 1; at Visit 4, pre-dose (within 30 min prior to the dose) and 5 hours (± 30 min) post-dose.

For participants who complete the 24-week Treatment Period and are entering the open-label Study PHA022121-C307 on the same day, Visit 6 PK sample collection serves as pre-dose Day 1 of Study PHA022121-C307.

Pharmacokinetic sample collection is summarized in Table 5.

Collected samples will be stored as specified in the Laboratory Manual. Plasma samples will be transferred frozen to the bioanalytical laboratory contracted by the Sponsor.

Plasma will be analyzed to determine concentrations of deucrictibant and metabolites using validated LC-MS/MS methods under the supervision of the Sponsor.

12.8.3. Optional Biomarker Sampling

Blood samples for potential assessment of biomarkers related to HAE or deucrictibant will be collected during the Screening Visit and Visit 6 or the EOT Visit (whichever comes first) with consent from the participant. Consent to provide these samples is optional. These samples will be stored for testing of biomarkers and will be destroyed within 5 years after completion of the study. The samples will not be collected in adolescents.

12.9. Adverse Events and Serious Adverse Events

12.9.1. Definition of Adverse Events

12.9.1.1. Adverse Event

An AE is defined as any untoward medical occurrence in a patient or clinical investigation participant administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment. An AE can therefore be any unfavorable and unintended sign, symptom, or disease temporally associated with the use of a study drug, whether or not considered related to the study drug.

Baseline medical history/conditions, abnormal physical examination findings, or diagnostic/laboratory values present at the Screening Visit are not considered to be AEs and should be recorded in the eCRF as baseline medical history. Events with a start date after the signing of the ICF are considered AEs, and any worsening in severity, or an increase in frequency, or change of concomitant medication for management of a pre-existing baseline medical condition is considered to be an AE. For the purpose of this protocol, HAE attacks and their associated symptoms will not be defined as AEs, unless they meet the criteria for an SAE. Prescheduled hospitalizations for non-emergency procedures will not be considered SAEs.

12.9.1.2. Serious Adverse Event

An SAE or reaction is any untoward medical occurrence that at any dose:

- Results in death
- Is life-threatening (at the time of the event)

  NOTE: The term “life-threatening” in the definition of “serious” refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.

- Requires inpatient hospitalization or prolongation of an existing hospitalization (note: a pre-existing condition that requires planned hospitalization is not considered as SAE)
  - Hospitalization for elective treatment of a pre-existing condition (ie, existed before signing the ICF) that did not worsen from baseline is not considered a criterion of seriousness, per se. Elective hospitalizations will therefore be reviewed for seriousness by the Sponsor on a case-by-case basis, and with Investigator input.
- Results in persistent or significant disability/incapacity
- Is a congenital anomaly/birth defect
- Is an important medical event that may jeopardize the participant or may require intervention to prevent one of the other outcomes listed above

Important medical events that may not result in death, be life threatening, or require hospitalization may be considered an SAE when, based upon appropriate medical judgment, they may jeopardize the participant or may require medical or surgical intervention to prevent one of the outcomes listed in the definition above.

12.9.1.3. Adverse Events of Special Interest

An AESI (serious or non-serious) is one of scientific and medical concern specific to the Sponsor's product or program, for which ongoing monitoring and rapid communication by the Investigator to the Sponsor may be appropriate. Such an event may require further investigation in order to characterize and understand it.

Instructions on reporting AESIs are provided in Section 12.9.5.3.

12.9.1.3.1. Neurological Adverse Events

While the neurological effects in nonclinical studies are believed to be associated with supratherapeutic C_{max} systemic concentrations, neurological AEs will be monitored as adverse events of special interest (AESIs) in this study as a precautionary measure. All serious and non-serious neurological severe adverse events (Grade 3 or higher) will be reported to the Sponsor/contract research organization (CRO) using the SAE form within 24 hours following the SAE reporting procedure (see Section 12.9.5.1). This reporting procedure will facilitate prompt and accurate evaluation.

12.9.1.4. Clinically Significant Abnormalities

Clinically significant abnormalities, including laboratory test results, vital signs, and ECG parameters, should be reported as an AE unless it is a component of a reached diagnosis, in which case the reached diagnosis is the AE. Clinically significant abnormalities are defined as abnormalities that require investigation and/or treatment. Repeating an abnormal test is not considered sufficient for clinical significance.

12.9.2. Relationship to Study Drug

All AEs/SAEs (also referred to as "events") will be assessed to determine their relationship to the study drug (causality). The Investigator must assess the causality using the binary choices of "Related" or "Not Related," and by considering the rationales listed below. Events should be considered "Related" only if there is a reasonable possibility of a causal relationship to the study drug. These rationales allow the Investigator to more clearly support their own assessments. All AEs and SAEs should be reported, regardless of causality, including events that are thought to be "Related" to targeted disease and/or population.

Examples of rationales for a **Not Related** assessment may include:

- Event attributed to concomitant medication or disease
- Event not reasonably temporally associated with study drug administration
- Negative dechallenge and/or negative rechallenge

Examples of rationales for a **Related** assessment may include:

- Temporal relationship of event to study drug exposure
- Event is known to be associated with the study drug or drug class
- Event improved on discontinuation of study drug
- Event reoccurred on rechallenge with study drug
- Event cannot be explained by disease or other drugs
- Biological plausibility

12.9.3. Recording Adverse Events

The Investigator or designee will be responsible for assessing AEs and SAEs. Reporting of AEs and SAEs (events) may result from spontaneous reports by the participant, general questioning by the Investigator or designee, clinically significant abnormalities found during physical examination, or clinically significant findings from vital signs, ECGs, or diagnostic test results including laboratory tests. When assessing potential AEs, questioning should be open-ended, without leading the participant.

The Investigator is responsible for monitoring the safety of participants and for providing appropriate medical care. In addition, the Investigator is responsible for alerting the Sponsor or its designee to any AEs, SAEs, or findings that seem unusual (eg, an abnormal laboratory test result), regardless of its relationship to study drug, and even if it may be considered to be an unanticipated beneficial effect.

AEs and SAEs should be described in acceptable medical terminology using a single reached diagnosis (versus signs and symptoms of the reached diagnosis). Abnormal diagnostic test results that support and are a component of the reached diagnosis should not be recorded (use the reached diagnosis). However, if a diagnosis is initially unknown or uncertain, any signs and symptoms (or diagnostic test results) must be recorded. As soon as diagnosis is known, the event term(s) must be adjusted to the final reached diagnosis.

Other event information to be collected includes (but is not limited to): onset date of the event, assessment of Common Terminology Criteria for Adverse Events (CTCAE) grade/severity, relationship to study drug (ie, causality assessment), action taken with study drug, outcome of the event, and date of resolution/stabilization of the event.

See Section 12.9.4 for information regarding follow-up for AEs and SAEs.

The Sponsor/designee retains the right to request additional information later for any participant with ongoing AEs/SAEs that are present at the End of Study Visit, if judged necessary.

12.9.3.1. Timing

All AEs and SAEs are to be recorded in the eCRFs from the time of signing informed consent and/or assent until the End of Study Visit. In certain situations, AE and SAE information is collected after the End of Study Visit (as outlined in Section 12.9.4).

AEs occurring after the participant's first dose of study drug will be considered TEAEs.

AEs occurring prior to the first dose of study drug that are ongoing and worsen in severity after the first dose of study drug are to be entered in the eCRF as a new AE, with start date as the day the AE severity worsened.

12.9.3.2. Severity

The National Cancer Institute (NCI) CTCAE version 5.0 (NCI-CTCAE, 2017) is a descriptive terminology which can be utilized for event/AE reporting. Adverse events (also referred to as "events") are classified by CTCAE grade/severity.

The CTCAE incorporates certain elements of the Medical Dictionary for Regulatory Activities (MedDRA) terminology. The System Organ Class (SOC), is the highest level of the MedDRA hierarchy, is identified by anatomical or physiological system, etiology, or purpose (eg, the SOC called "Investigations" is used for laboratory test results). An AE is a term that is a unique representation of a specific event used for medical documentation and scientific analyses. Each CTCAE term is a MedDRA lowest level term.

Per NCI CTCAE, grade refers to the severity of the event. The CTCAE grade/severity should be evaluated and recorded on the eCRF according to the NCI CTCAE, version 5.0 (NCI-CTCAE, 2017).

If an AE description is not graded according to the CTCAE grading system, the Investigator will assess the severity of AEs based on the following CTCAE grading definitions:

- Grade 1: Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
- Grade 2: Moderate; minimal, local or noninvasive intervention indicated; limiting age- appropriate instrumental activities of daily living (ADL)*.
- Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL**.
- Grade 4: Life-threatening consequences; urgent intervention indicated

- Grade 5: Death related to AE
  - Not all Grades are appropriate for all AEs. Therefore, some AEs are listed with fewer than five options for Grade selection.

* Instrumental ADL refers to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

** Self-care ADL refers to bathing, dressing, and undressing, feeding self, using the toilet, taking medications, and not bedridden.

If the severity of the AE changes over the course of the AE, the highest level of severity will be recorded in the eCRF.

12.9.4. Adverse Event Follow-up

AEs should be followed until the event resolves or until the Investigator assesses it as chronic or stable (ie, resolved with sequelae), the participant withdraws from the study, the participant is lost to follow-up, or the participant dies. For AEs that are ongoing or resolving at the time of EOS Visit, attempts will be made by the Investigator to collect follow-up information, in particular for all SAEs and all Grade ≥3 AEs.

AE data will be recorded in the eCRF up to the time of database lock, prior to which the status of all outstanding ongoing AEs will be reviewed and assessed by Sponsor or designee. As considered appropriate by the Sponsor or designee, follow-up information will continue to be required from the Investigator on specific AEs (and in any case all SAEs considered related to the study drug), although related follow-up details may not be recorded in the CRF after database lock.

As per regulations, if the Investigator becomes aware of an SAE after the EOS Visit and the SAE is considered by the Investigator to be at least possibly related to the study drug, the SAE will be reported to the Sponsor in accordance with Section 12.9.5.1.

12.9.5. Reporting Adverse Events

12.9.5.1. Reporting Serious Adverse Events

All SAEs after signing of the informed consent through the EOS Visit must be reported to the Sponsor/designee by the Investigator within 24 hours of the Investigator's knowledge of their occurrence regardless of causality.

The SAE must be reported by immediately completing the appropriate fields of the eCRFs using acceptable medical terminology, written in English, and must include a causality assessment (relationship to study drug). Simultaneously, any other applicable eCRF pages, including relevant AE and concomitant medications eCRF pages, must also be completed. Supporting source documents such as hospital discharge summaries may be provided via the SAE reporting contact information provided below.

If the electronic data capture (EDC) system is unable to be accessed, the SAE Reporting contact information listed below should be utilized. A paper SAE Report Form should be completed and sent within 24 hours. The paper SAE report form should be sent by e-mail or fax to:

Medpace Clinical Safety	
SAE Reporting Telephone Line:	+1-800-730-5779, dial 3 or +1-513-579-9911, dial 3 (the Americas) +49 89 89 55 718 44 (EU) +81-3-4563-7000, dial 3 (Asia Pacific)
Safety Fax:	+1-866-336-5320 or +1-513-570-5196 (the Americas) +49 89 89 55 718 104 (EU)
Email:	medpace-safetynotification@medpace.com

If a paper SAE Report Form was utilized, the original paper form, together with the fax confirmation sheet (if applicable), must be retained at the study site. If the initial SAE report is made verbally, via email, or by telephone, a written form of the SAE must follow within 24 hours (via eCRF entry or paper SAE Report Form). The Investigator will be requested to provide detailed information regarding the event at the time of the initial report.

Each SAE should be recorded as a single reached diagnosis on the SAE report form and AE eCRF. Accompanying signs or symptoms (eg, physical examination findings, abnormal diagnostics test results, ECG findings) should not be reported as additional SAEs. However, if initially the diagnosis is unknown, the signs and symptoms should be reported. As soon as a reached diagnosis causing the signs and symptoms is known, the event term(s) should be amended to the final reached diagnosis. If there are multiple signs and symptoms, all related signs and symptoms should be collapsed into the final reached diagnosis, and this will require inactivation of previously reported AEs.

For all SAEs occurring during the study, the Investigator must submit follow-up reports to the Sponsor or its designee regarding the participant's subsequent course until the SAE has resolved, or until the condition is chronic or stabile (resolved with sequelae), the participant withdraws from the study, the participant is lost to follow-up, or the participant dies. All follow-up information should be reported to the Sponsor/designee within 24 hours of awareness.

The Sponsor's Medical Lead will ultimately be responsible for determining whether an SAE is expected or unexpected utilizing the current reference safety information that is in place at the time of the event's occurrence and assessment. An SAE will be considered unexpected if the nature, grade/severity, or frequency of the event is not consistent with the list of expected adverse reactions.

12.9.5.2. Reporting Suspected Unexpected Serious Adverse Reactions

Following notification from the Investigator, the Sponsor and/or designee (ie, CRO) will report AEs which are both serious and unexpected (ie, not previously described in the reference safety information in the approved IB) and that are assessed as related to the study drug to the FDA, European Medicines Agency (EMA)/EudraVigilance, other concerned Regulatory Authorities, IRB/IECs and Investigators within the required timelines as specified in 21 Code of Federal Regulations (CFR) Part 312.32, Regulation (EU) No. 536/2014, the European Commission's "Detailed guidance on the collection, verification, and presentation of adverse event/reaction reports arising from clinical trials on medicinal products for human use" (CT-3, June 2011), ICH GCP Guidelines, and other applicable national regulatory requirements. Typically, fatal and life-threatening adverse reactions must be reported immediately and not later than 7 calendar

days and all other unexpected serious adverse reactions must be notified immediately and no later than 15 calendar days. All other SAEs whether related to the study drug will be reported to the concerned Regulatory Authorities, and IRB/IECs at least annually in a summary format (eg, Development Safety Update Report).

Additionally, events may occur during the clinical development of a study drug which do not fall within the definition of a suspected unexpected serious adverse reaction (SUSAR) and thus are not subject to the reporting requirements for SUSARs, even though they may be relevant in terms of participant safety. Examples are new events (or findings) related to the conduct of a study or the development of a study likely to affect the safety of participants such as:

- A significant hazard to the participant population such as lack of efficacy of a study drug used for the treatment of a life-threatening disease
- A major safety finding from a newly completed animal study (such as carcinogenicity)
- A temporary halt of a study for safety reasons if the study is conducted with the same study drug in another country by the same Sponsor, where relevant for the safety of participants

These events (or findings) are not to be reported as SUSARs, but they might require other action, such as urgent safety measures and their notification, substantial amendments, or early termination of the study, and will be reported in accordance with applicable local regulations and guidelines.

12.9.5.3. Reporting Adverse Events of Special Interest

All serious or severe neurological adverse events (Grade 3 or higher) will be reported to the Sponsor/CRO using the SAE form within 24 hours following the SAE reporting procedure (Section 12.9.5.1). This reporting procedure will facilitate prompt and accurate evaluation.

12.10. Reporting of Pregnancy

Any female participant who has taken study drug and who becomes pregnant during the study will immediately discontinue the study. The pregnancy will be followed until the pregnancy's outcome to assess for any potential occurrence of congenital anomalies or birth defects (discovered in utero or at birth). After resolution of the pregnancy, the participant may enroll in the long-term extension study, provided all entrance criteria are met for that study.

While pregnancy is not considered an AE per se, pregnancies must be reported immediately to the Sponsor or designee using the Pregnancy Report Form and following SAE reporting timelines (ie, reporting from the Investigator/site to the Sponsor/designee) in Section 12.9.3.1.

Pregnancy outcomes will be reported on the Pregnancy Outcome Report Form. Pregnancy outcomes include live birth (full term or pre-term), stillbirth, spontaneous abortion, and induced abortion (therapeutic or elective).

If a spontaneous abortion occurs or a congenital abnormality/birth defect is observed (including in utero detections), they must be reported immediately to the Sponsor or designee.

Some pregnancy findings and outcomes (eg, spontaneous abortion, stillbirth, congenital abnormality/birth defect [including in utero detections]) may meet seriousness criteria; if so, the SAE should also be reported.

12.11. Reporting of Study Treatment Errors, Including Misuse/Abuse

Medication errors are unintentional errors in the prescribing, dispensing, administration, or monitoring of a medicine or study drug while under the control of a healthcare professional, participant, or consumer. The administration or consumption of an unassigned treatment and administration of an expired product are always reportable as medication errors; cases of patients missing doses of investigational product are not considered reportable as medication errors.

In a clinical study, misuse refers to situations where the study drug is intentionally and inappropriately used not in accordance with the protocol.

In a clinical study, abuse corresponds to the persistent or sporadic, intentional excessive use of a study drug, which is accompanied by harmful physical or psychological effects.

In this clinical study, overdose refers to situations where the participant ingests more than 1 tablet of study drug within 18 (24 minus 6) hours of the previous dose.

Special situations, including any unintended study treatment/medication error, misuse, abuse, or overdose of study drug described above must be reported on the Special Situations Report Form and sent to Medpace Clinical Safety within 24 hours of Investigator's awareness.

If any special situation as described above is associated with/results in an AE or SAE, then the AE or SAE must be recorded in the eCRF within the same 24-hour reporting timelines. Refer to Section 12.9.5 for SAE reporting procedures.

13. STATISTICS

13.1. General Considerations

Descriptive summary statistics will be presented for all study endpoints.

For continuous endpoints, descriptive summary statistics will generally include number of observations, mean, standard deviation (SD), median, interquartile range (Q1, Q3), and range (minimum, maximum).

For categorical endpoints, summaries will generally include numbers and percentage by category.

Formal statistical testing will be performed on the primary, secondary efficacy, and PRO endpoints. A gatekeeping procedure will be used to control the study-wise type 1 error rate of 5% (2-sided) in testing the primary, secondary efficacy and PRO endpoints in the fixed order as presented in Section 13.11.

Further details about all planned analyses will be provided in the study Statistical Analysis Plan (SAP), which will be finalized before the database lock and will include a more technical and detailed description of all analyses.

The statistical software, SAS® version 9.4, or higher, will be used for all summaries and statistical analyses.

13.2. Determination of Sample Size

A Poisson distribution of Investigator-confirmed HAE attacks was assumed for the analysis of primary endpoint 'Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment period'. In the Phase 2 study PHA022121-C301, the [redacted] in the placebo group.

With a conservative assumption that the time-normalized (4 weeks) number of Investigator-confirmed HAE attacks [redacted] in the placebo group during the 24-week Treatment Period, a sample size of 81 randomized participants (2:1 randomization; 54 on deucrictibant 40 mg XR tablet and 27 on placebo) will provide a power of at least 95% to detect a 50% reduction in time-normalized number of Investigator-confirmed HAE attacks for deucrictibant 40 mg XR tablet comparing to placebo, based on a 2-sided test at a significance level of 5%.

13.3. Analysis Sets

13.3.1. Intent-to-Treat (ITT) Analysis Set

The ITT Analysis Set includes all randomized participants who received at least one dose of study drug (deucrictibant 40 mg XR tablet or placebo). In the analyses performed on the ITT Analysis Set, participants will be analyzed according to their assigned study drug per randomization regardless of actual study drug received.

The ITT Analysis Set is the primary analysis set within the study.

13.3.2. Per Protocol (PP) Analysis Set

The PP Analysis Set includes all participants in the ITT Analysis Set without any major protocol violations that could influence the validity of the efficacy evaluations.

Major protocol violations that may impact the efficacy assessments include, but are not limited to:

- Major violations of inclusion/ exclusion criteria
- Actual study drug is different from assigned at randomization
- Use of concomitant prohibited medications during the 24-week Treatment Period

All criteria to exclude participants from the PP Analysis Set will be finalized based on a blinded data review meeting prior to database lock.

13.3.3. Safety Analysis Set

The Safety Analysis Set includes all randomized participants who received at least one dose of study drug (deucrictibant 40 mg XR tablet or placebo). The Safety Analysis Set will be analyzed according to the participants' actual study drug received.

13.3.4. Pharmacokinetic Analysis Set

The PK Analysis Set includes all randomized participants who received at least one dose of study drug (deucrictibant 40 mg XR tablet or placebo) and undergo plasma PK sampling and have evaluable concentration data for analysis.

13.4. Participant Disposition and Baseline Characteristics

Participant disposition will be summarized using all participants, including information of the Screening, Run-in, 24-week Treatment, and Follow-up Periods. Reasons for discontinuation of study drug and withdrawal from study will be summarized and presented in listings.

Baseline characteristics including demographics, HAE disease history, medication history and stratification factors will be summarized by treatment group for the ITT Analysis set.

13.5. Analyses of Efficacy

13.5.1. Primary Endpoint, Estimand, and Analysis

The primary endpoint of this study is time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168).

The primary estimand of interest is defined by the following attributes:

Treatment	Deucrictibant 40 mg XR tablet versus placebo, orally administered QD
Population	The target patient population is defined through the study inclusion/ exclusion criteria
Endpoint	Time normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day 1 through Day 168), also referred to as the 4-week HAE attack rate, or HAE attack rate, which is calculated for each participant as number of Investigator-confirmed HAE attacks occurring during the 24-week Treatment Period divided by the number of days the participant contributed to the 24-week Treatment Period, multiplied by 28 days.
Intercurrent events (ICEs) and strategies to handle ICEs	a) Early study drug discontinuation for any reason unless accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period b) Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period c) Use of prohibited medication (including prohibited short/long-term HAE prophylaxis) during the 24-week Treatment Period d) Non-compliance to study drug during the 24-week Treatment Period *For ICE b), the composite variable strategy will be applied.* *For ICEs a), c), and d), the treatment policy strategy will be applied.*

Population-level summary	Ratio of the 4-week HAE attack rates between deucrictibant 40 mg XR tablet and placebo

Abbreviations: HAE = hereditary angioedema; ICE = intercurrent event; LTP = long-term prophylaxis; QD = daily; XR = extended- release.

Composite variable strategy will be used to address the ICE 'Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period'. A poor value of HAE attack rate will be assumed after occurrence of such ICE through the end of the 24-week Treatment Period.

For the remaining ICEs, a treatment policy strategy will be applied. This means that all data, regardless of the occurrence of the ICE, will be included in the analysis.

For the primary endpoint, the following null hypothesis and alternative will be tested for deucrictibant 40 mg XR tablet against placebo:

- H0: $\lambda_{\text{deucrictibant 40 mg XR}} / \lambda_{\text{placebo}} = 1$
- H1: $\lambda_{\text{deucrictibant 40 mg XR}} / \lambda_{\text{placebo}} \neq 1$

$\lambda_{\text{deucrictibant 40 mg XR}}$ refers to the time normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period in deucrictibant 40 mg XR tablet treatment group.

λ_{placebo} refers to the time normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period in placebo treatment group.

The **null hypothesis H0** is that the Investigator-confirmed HAE attacks rate ratio = 1, which means that there is no difference between treatment groups.

The **alternative H1** is that the Investigator-confirmed HAE attacks rate ratio ≠ 1, which means that there is difference between treatment groups.

The analysis of the primary endpoint will be performed based on the ITT Analysis Set and will compare deucrictibant 40 mg XR tablet with placebo using a generalized linear model (GLM) assuming a Poisson distribution with a log link function and Pearson chi-square scaling of standard errors to account for potential (over)-dispersion. The model will include fixed effects for treatment group, the baseline attack rate and age group (adolescent and adult), and an offset term of the logarithm of participant's follow-up time during the Treatment Period.

The estimated least squares (LS) rate for deucrictibant 40 mg XR tablet and placebo and the rate ratio relative to placebo will be provided with 95% confidence intervals. These estimates will be reported as 4-week attack rates by transforming the estimates using the exponential function and scaling by the unit of time. The treatment effect in the primary efficacy endpoint evaluation will be tested using a Wald-based chi-square test.

13.5.1.1. Sensitivity Analysis for Missing Data

Sensitivity analyses for handling missing data will include pattern mixture model by multiple imputation and tipping point analyses.

13.5.1.2. Supplementary Analyses

The following supplementary analyses will be implemented:

- Supplementary analysis based on the Per-Protocol (PP) Analysis Set
- Supplementary analysis using a negative binomial regression (instead of a Poisson regression)
- Supplementary analysis using a Wilcoxon two-sample test. The difference between the treatment groups will be based on the Hodges-Lehman estimate.

13.5.2. Secondary Efficacy Endpoints, Estimands, and Analyses

Secondary efficacy endpoints will be analyzed based on the ITT Analysis Set.

For continuous endpoints, the estimand framework and analysis method for the following secondary efficacy endpoints are defined similarly as for the primary endpoint:

- Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks treated with on-demand medication during the 24-week Treatment Period
- Time-normalized (per 4 weeks) number of Investigator-confirmed moderate or severe HAE attacks during the 24-week Treatment Period
- Time-normalized (per 4 weeks) number of Investigator-confirmed severe HAE attacks during the 24-week Treatment Period

The binary endpoints are:

- Proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants achieving ≥70% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants achieving ≥90% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
- Proportion of participants that are HAE attack-free during the 24-week Treatment Period

The estimand framework for the binary endpoints above is presented below (example: ‘proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period’).

Treatment	Deucrictibant 40 mg XR tablet versus placebo, orally administered QD
Population	The patient population is defined through the study inclusion/ exclusion criteria who have been treated with study drug for at least 4 weeks.
Endpoint	Proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
Intercurrent events (ICEs) and strategies to handle ICEs	a) Early study drug discontinuation for any reason unless accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period

	b) Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period c) Use of prohibited medication (including prohibited short/long-term HAE prophylaxis) during the 24-week Treatment Period d) Non-compliance to study drug during the 24-week Treatment Period *For ICE b), the composite variable strategy will be applied.* *For ICEs a), c), and d), the treatment policy strategy will be applied.*
Population-level summary	Odds ratio of responder between deucrictibant 40 mg XR tablet and placebo. A responder is defined as a participant achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period. A responder (and non-responder) is derived based on the time normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period.

Abbreviations: HAE = hereditary angioedema; ICE = intercurrent event; LTP = long-term prophylaxis; QD = daily; XR = extended -release.

Composite variable strategy will be used to address the ICE 'Early study drug discontinuation for any reason accompanied by a switch to any alternative LTP HAE therapy during the 24-week Treatment Period'. Following the composite variable strategy, a participant with this ICE will be regarded as not achieving ≥50% reduction in HAE attack rate, ie, a non-responder, in the analysis.

For the remaining ICEs, a treatment policy strategy will be applied. This means that all data, regardless of the occurrence of the ICE, will be included to determine whether the participant is a responder, namely, whether the participant has achieved ≥50% reduction in HAE attack rate relative to baseline in the 24-week Treatment Period.

For each treatment group, the number and the proportion of responders and non-responders with corresponding 95% CI will be presented. A logistic regression model including fixed effects for treatment group, the baseline attack rate and age group (adolescent and adult) will be used to assess the difference between the treatment groups.

The proportion of time without angioedema symptoms during the 24-week Treatment Period will be summarized using descriptive statistics by treatment group. The proportions between treatment groups will be compared using an ANCOVA model including fixed effects for treatment group, the baseline attack rate and age group (adolescent and adult).

13.6. Analysis of Safety

The analysis of safety and tolerability will be performed on the Safety Analysis Set.

Treatment-emergent AEs (TEAEs) are defined as AEs that start or pre-existing AEs that have worsened after the initiation of study drug.

The number and percentage of participants with TEAEs will be summarized by treatment group by body SOC and PT using the Medical Dictionary for Regulatory Activities (MedDRA®, version 27.0 or later).

Summaries in terms of severity and relationship to study drug will also be provided.

AESIs, treatment-emergent serious adverse events (TESAEs), and TEAEs leading to study drug discontinuation will be summarized in a similar fashion.

Listings of AEs, SAEs, and AEs leading to study drug discontinuation will also be provided.

Any prior and concomitant medication (including standard-of-care treatment to manage HAE attacks) will be coded using the World Health Organization (WHO) medication dictionary (WHODD GLOBAL version B3 March 2024, or later). Prior and concomitant medication and procedures will be summarized by treatment group.

Laboratory data will be presented in summary tables of observed values and change from baseline values, and by presented in shift tables. Individual listings will also be provided.

Vital signs, physical examination, and ECG data will be presented in summary tables and individual listings by treatment group.

13.7. Analysis of Patient Reported Outcomes

AE-QoL, AECT-4wk, PGA-Change, WPAI-SHP and TSQM-9 will be summarized for each visit by treatment group.

The change from baseline in:

- AE-QoL total score
- AE-QoL functioning domain score
- AE-QoL fears/ shame domain score

at Week 24 will be analyzed using a mixed-effect model with repeated measures (MMRM), based on this model the treatment effect at Week 24 will be estimated. LS means, and the 95% confidence intervals will be presented.

The response variable in the model will be the change from baseline at each scheduled post-randomization visit up to and including Week 24. Treatment group, visit, age group, baseline HAE attack rate and treatment by visit interaction will be included as factors in the model. Baseline of the corresponding AE-QoL score will be also included in the model as a continuous linear covariate.

The analysis of AECT 4-wk at Week 24 will be based on the categories presented in Section 11.3: a total score of 0 to 9 points indicates poorly controlled disease and 10 to16 points indicates well controlled disease. The analysis will be based on a logistic regression model similar to that specified in Section 13.5.2.

13.8. Pharmacokinetics Analysis

The plasma concentrations and metabolites of deucrictibant 40◦mg XR tablet will be summarized.

The PK evaluation will consist of a population (pop) PK analysis of the samples taken in this study.

13.9. Other Analyses

Not applicable.

13.10. Interim Analysis

No interim analysis is planned for this study.

13.11. Multiplicity

Formal statistical testing will be performed on the primary, secondary efficacy and PRO endpoints.

A gatekeeping procedure will be used to control study-wise type 1 error rate at a 2-sided alpha of 0.05 across all endpoints tested, following the order the primary endpoint first, followed by the secondary efficacy and PRO endpoints in the below fixed sequence:

1. Time-normalized (per 4 weeks) number of Investigator-confirmed HAE attacks during the 24-week Treatment Period (Day Day 1 through Day Day 168)
2. Time-normalized number of Investigator-confirmed HAE attacks treated with on-demand medication during the 24-week Treatment Period
3. Time-normalized number of Investigator-confirmed moderate or severe HAE attacks during the 24-week Treatment Period
4. Proportion of participants achieving ≥50% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
5. Proportion of participants achieving ≥70% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
6. Proportion of participants achieving ≥90% reduction in HAE attack rate relative to baseline during the 24-week Treatment Period
7. Proportion of participants that are HAE attack-free during the 24-week Treatment Period
8. Time-normalized number of Investigator-confirmed severe HAE attacks during the 24-week Treatment Period
9. Proportion of time without angioedema symptoms during the 24-week Treatment Period
10. Change from baseline in AE-QoL total score at Week 24
11. Change from baseline in AE-QoL functioning domain score at Week 24
12. Change from baseline in AE-QoL fears/ shame domain score at Week 24
13. Proportion of participants that are well controlled on AECT-4wk at Week 24

14. DIRECT ACCESS TO SOURCE DATA/DOCUMENTS

14.1. Study Monitoring

The Sponsor is responsible for monitoring the clinical study to ensure that participants' human rights, safety, and well-being are protected, that the study is properly conducted in adherence to the current protocol and any study-specific documents (eg, eCRF completion guidelines), to GCP ICH E6, and to applicable regulatory requirements, and that study data reported by the Investigator are accurate and complete and that they are verifiable with study-related records such as source documents. No data that does not have source documentation will be recorded in the eCRF. The Sponsor or designee will determine the extent and nature of the monitoring and assign sponsor-independent clinical monitors who are GCP-trained. The Sponsor or designee will provide the monitors with adequate training on the study protocol. The study will be monitored in accordance with the clinical study monitoring plan, which specifies the monitoring tasks.

Monitoring visits will be performed to confirm appropriate study conduct. If applicable and permissible per local regulations, monitoring visits may be performed remotely. Monitoring or remote monitoring may include assessment of the following:

- The investigational team is adhering to the study protocol
- Informed consent has been obtained from all participants
- AEs/SAEs are being reported as required
- Data are accurately recorded in the eCRFs
- Study drug is being stored correctly and drug accountability is being performed on an on-going basis
- Facilities are, and remain, acceptable throughout the study
- The Investigator and site are receiving sufficient information and support throughout the study
- eCRFs and Study Records

A validated EDC system will be used for entry of the data by the Investigator or other authorized user into study specific eCRFs designed by the Sponsor or its designee. All information recorded on eCRFs for this study must be consistent with the participant's source documentation. The Investigator or designee will attempt to obtain specific source documents for all data recorded in the eCRF (eg, eligibility criteria, medical history). Where there is no specific source available in the medical records, the Investigator or designee will attempt to obtain specific, written source documents from previous treating physicians. Only when no written source is available may the Investigator or designee obtain responses from the participant him/herself during the intake interview(s), record these into the source documents, and enter the data into the eCRF.

Initial data entry and any changes to the data will be made only by authorized users, and data entries and changes will be captured in an electronic audit trail. An explanation of any data change will be recorded.

14.2. Audits and Inspections

Authorized representatives of the Sponsor, a regulatory authority, an IEC, or an IRB may visit the site to perform audits or inspections, including source data verification. The purpose of a Sponsor audit or inspection is to systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, GCP guidelines of the ICH, and any applicable regulatory requirements. The Investigator should contact the Sponsor or designee immediately if contacted by a regulatory agency about an inspection.

14.3. Independent Ethics Committee/Institutional Review Board

In accordance with GCP, the protocol, ICF(s), recruitment materials, and all participant materials will be submitted to an appropriate IEC/IRB for review and approval. Approval of both the protocol and the ICF must be obtained before any participant is enrolled. Any substantial amendment to the protocol will require review and approval by the IEC/IRB before the changes are implemented to the study. Non-substantial amendments, eg, containing administrative changes, will only be submitted to the IEC/IRB for notification. All changes to the consent form must be IEC/IRB approved, in accordance with local regulations.

14.4. Independent Data Monitoring Committee

An IDMC will be assembled before the start of the study to review safety data on a regular basis (and on an ad hoc basis in case of unexpected safety issues) for the study. The IDMC will serve for deucrictibant prophylactic treatment studies and review safety data of both Studies PHA022121-C305 and PHA022121-C307. The Charter will be prepared before the start of the study and will identify the IDMC membership and outline the expected review of unblinded safety data. The IDMC will consist of independent experts external to the study.

The responsibilities of the IDMC include, but are not limited to:

- To minimize the exposure of participants to an unsafe therapy or dose
- To make recommendations for changes in study processes where appropriate
- To make recommendations on study continuation

Further details on the membership, responsibilities and working procedures of the IDMC are described in the IDMC Charter, which is provided as a separate document in the study file.

15. QUALITY CONTROL AND QUALITY ASSURANCE

Procedures relevant to the clinical management of this study are either described in the applicable standard operating procedures (SOPs) of the Sponsor or their representatives. In accordance with the Sponsor's SOPs, this study may be subject to independent quality assurance (QA) GCP audits at the study sites.

The Sponsor will not allow any waivers to the protocol throughout the conduct of the study. A deviation from the protocol (not implemented in an amendment or intended as a systematic

change) that could not be justified as necessary to protect the safety, rights, or welfare of the participants will be reported and followed up according to the local IEC/IRB requirements.

Investigators and other study personnel will be appropriately trained by the Sponsor or its designee. The study will be monitored (see also Section 14.1).

To ensure the collection of accurate, consistent, and reliable data, safety laboratory tests will be done at a central laboratory. Based on Investigator or designee determination or local requirement for participant safety, additional samples may be assessed during the study by a local laboratory, in parallel with the central laboratory.

Serious breaches will be reported in accordance with EU/European Economic Area Regulation (EU) No 536/2014, Article 52:

- The Sponsor or designee shall notify the Member States concerned about a serious breach or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than 7 days of becoming aware of that breach.

A serious breach is defined as a breach likely to affect to a significant degree the safety and rights of a participant or the reliability and robustness of the data generated in the clinical study.

16. ETHICS

16.1. Ethical Conduct of the Study

The study will be conducted in accordance with this study protocol, the ICH GCP (ICH GCP E6), 536/2014 EU CTR, and other applicable local laws and regulations. Compliance with ICH-GCP provides public assurance that the rights, safety, and well-being of study participants are protected, consistent with the principles that have their origin in the Declaration of Helsinki, and that the clinical study data are credible.

The Investigator will assure that no deviation from, or changes to, the protocol will take place without prior agreement from the Sponsor and documented approval from the IEC/IRB, except where necessary to eliminate an immediate hazard to the participants. All relevant personnel involved in the conduct of this study will have completed ICH GCP training.

Participant insurance will be provided according to local regulations.

16.2. Written Informed Consent

At the Screening Visit, a signed ICF/assent must be obtained from each participant and/or parent/legally designated representative/guardian before performing any study-related procedures. Each participant will be given both verbal and written information describing the nature and duration of the clinical study.

The ICF must describe in detail the study drug, study procedures, and risks/potential burden of study participation. The ICF and any other written study material provided to the participant must be approved by an IEC/IRB before participant enrollment.

The Investigator or designee is responsible for explaining the nature and purpose of the study as well as other study-related matters to participants, using the written information, and for obtaining their full understanding and written consent to participate in the study at their own free will. No participant may be subjected to undue influence, such as compulsory enrollment into a study. The Investigator or designee who provided explanations to the participant should sign and date the ICF as well as the participant. The participant should receive a copy of the signed ICF (or a duplicate original if required by local regulations) and an original must be filed by the Investigator. The process and communication of the consent should be documented on medical records of the participant. The signed ICFs will be retained by the Investigator and made available upon request (for review only) to the study monitor and auditor.

The language and expressions used in the written information should be as plain and understandable as possible. Participants should be given the opportunity to ask questions and receive satisfactory answers to the inquiry and should have adequate time to decide whether or not to participate in the study. Written information should not contain any language or content that causes the participant to waive or appears to waive any legal rights, or that releases/mitigates or appears to release/mitigate the study site, the Investigator, other study personnel, collaborators, or the Sponsor from liability for negligence.

For adolescent participants (≥12 to <18 years of age) the language and expressions used in the written assent and other information should be age appropriate and a separate written consent should be provided to the participant's parent/legally designated representative/guardian, in accordance with local regulations.

If an adolescent participant reaches the age of 18 years or as determined by local law during study participation, the participant will be required to provide written informed consent in order to continue study participation, in accordance with local regulations.

If an ICF is updated as a result of a substantial protocol amendment, the new IEC/IRB approved version will be used to re-consent currently enrolled participants and must be provided to newly enrolled participants before their entry into the study.

16.3. Participant Confidentiality

The conduct of this study and the processing of any personal data collected from each participant (or from a participant's healthcare professional or other relevant third-party sources) by the Sponsor or its designee, the site, and the Investigator for use in the study will fully adhere to the requirements set out in applicable data protection and medical privacy laws or regulations, including, without limitation, the General Data Protection Regulation (GDPR) EU 2016/679. The Sponsor or its designee shall ensure that at all times it has an appropriate legal basis for processing personal data under applicable data protection law. Site-based organizational and technical arrangements to avoid unauthorized access vary by site but all include access-controlled/access-limited document control and technical solutions including passwords and security control measures to protect study-specific data, both in paper and electronic format.

The Investigators shall provide coded data to the Sponsor or its designee, which does not reveal the participant's name, full date of birth, or any other information which can identify the participant. All personal information shall be replaced with a Subject Identification Code before any information leaves the Investigator sites.

The Investigator shall report any data breaches that might occur to the Sponsor or its designee without undue delay. The Sponsor has implemented a Business Practice to address Data Breaches that complies with the requirements of applicable laws and regulations including the GDPR. The Data Breach procedures in the Business Practice provide specific responses to actual or potential threats and involve investigation, containment, and mitigation. If applicable, the authorities and the data subjects shall be notified of a data breach, within the required timeframes of the applicable laws and regulations, including those of the GDPR.

The Sponsor will use the information obtained during the conduct of this study for the development of deucrictibant XR tablet. The Investigator is obliged to provide the Sponsor or its designee with complete test results and all data collected in this study, as described in this protocol.

Access to stored samples will be limited. Samples and data will be stored using codes assigned by the Sponsor. The code key linking the participant's name to the research records will be kept by the Investigator and may not be released outside the study site. The coded data does not include any identifying information for the participant, such as name or address. Data will be kept in password-protected computers. Only Investigators, the Sponsor and affiliates, or contracted parties will have access to the coded samples and data.

Data generated by this study must be available for inspection by representatives of the European Medicines Agency, the FDA, Health Canada, any other competent authority, and the IRB/IEC for the study sites. Even though individuals involved in the study, such as study monitors and auditors, may get to know matters related to participants' privacy due to direct access to source documents, or from other sources, they may not disclose these contents to third parties.

17. DATA HANDLING AND RECORDKEEPING

17.1. Inspection of Records

The Investigator and the study site must accept monitoring and auditing by the Sponsor or their representatives as well as inspections from the IEC/IRB and relevant regulatory authorities. Source documents may be reviewed remotely using regulatory-compliant and documented computerized systems when agreed upon with the Investigator and study site, as applicable by local regulations. In these instances, they must provide all study-related records, such as source documents when they are requested by the monitors and auditors, the IEC/IRB, or regulatory authorities. The confidentiality of the participants' identities shall be well protected consistent with local and national regulations when the source documents are subject to direct access. In accordance with the Sponsor's procedures, this study may be subject to an independent QA GCP audit at the study sites.

17.2. Retention of Records

All records (data and documents) relating to the study that are generated by or collected by the CRO will be filed in an electronic trial master file (eTMF) according to ICH-GCP. The eTMF will be available for Sponsor review at any time and will be transferred to the Sponsor after the study is complete and will be archived for at least 25 years or as otherwise required by local regulations. Also, all records relating to the study that are generated by or collected by the

Investigators will be filed in an Investigator site file according to ICH GCP. Investigators will retain the Investigator site file for at least 25 years or as otherwise required by local regulations. The Investigators will also retain source documents under their control for at least 25 years or as otherwise required by local regulations. All medical records will be retained by the medical center according to their own internal procedures.

17.3. Protocol Deviations

Protocol deviations will be defined and recorded based on the Protocol Deviation Plan. The following deviations from the protocol will be considered protocol violations:

- Deviations from the inclusion and exclusion criteria
- Deviations from concomitant medication restrictions
- Deviations from any other protocol requirements that result in a significant added risk to the participant or have an impact on the quality of the data collected or the outcome of the study.

Protocol deviations will be reported in the final Clinical Study Report.

17.4. Data Generation and Analysis

The Investigator or designee must enter all protocol-required data in the available eCRF, which is a validated system. In the interest of collecting data in the most efficient manner, the Investigator or designee should enter data into the eCRF as soon as possible after the participant's visit. Laboratory results from the central laboratory will be electronically transferred. The eCRFs and any supporting documents should be available for retrieval at any given time. The Monitor should verify the data in the eCRFs with source documents and confirm that there are no inconsistencies between them. If any inconsistency is detected on the eCRFs, the Monitor must query the Investigator and the Investigator or designee should make corrections/additions in the eCRF or provide an explanation within the eCRF system. The Monitor should verify the corrected data with source documents to confirm that there are no inconsistencies between them, and also check that appropriate records on the corrections/additions of data are maintained.

Electronic patient reported outcomes (ePRO) data will be collected with an electronic device and electronic application. The electronic device is a validated 21 CFR Part 11-compliant electronic data capture tool. Provisions will be made for back-up method(s) in case of device malfunction.

For Screening failures, defined as participants who signed informed consent but were not eligible for study participation, relevant data will be collected in the eCRF.

The Investigator is responsible for ensuring that all data in the eCRF are accurate and complete, and that all entries are verifiable with source documents. Source data must be available at the site to document the existence of the study participants, and to substantiate the integrity of study data collected. In case a participant is prematurely discontinued, the reason for discontinuation must be documented in the source documents and eCRF. Study data collected from a participant before study discontinuation will still be used in the overall data analysis.

Data management will be coordinated by a delegated CRO, in accordance with their SOPs for data management. All study specific processes and definitions will be described in the data management plan. Coding of medical terms will be performed using MedDRA and medications will be coded with the World Health Organization-Drug Dictionary.

17.5. Financial Disclosure

Any Investigator involved in conducting the study is required to disclose any financial arrangement during the study and for 1 year after, whereby the outcome of the study could be influenced by the value of the compensation for conducting the study, or other payments the Investigator received from the Sponsor. The following information is collected: any significant payments from the Sponsor or subsidiaries such as a grant to fund ongoing research, compensation in the form of equipment, retainer for ongoing consultation, or honoraria; any proprietary interest in the study drug; any significant equity interest in the Sponsor or subsidiaries as defined in 21 CFR 54 2(b). A financial disclosure form is to be signed by each Investigator.

18. PUBLICATION POLICY

All data generated from this study are the property of the Sponsor and shall be held in strict confidence along with all information provided by the Sponsor. Except as provided through written agreement between the Sponsor and the Investigator, independent analysis and/or publication of these data by the Investigator or any member of his/her staff is not permitted without prior written consent of the Sponsor. Such consent will not be withheld unreasonably. The Sponsor is in agreement with the principle of full disclosure of clinical study results.

Within one year from the end of a clinical study in all Member States concerned, the Sponsor will submit to the EU database CTIS a summary of the results of the clinical study. The content of the summary of the results is set out in CTR Annex IV. It shall be accompanied by a summary written in a manner that is understandable to laypersons.

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APPENDIX 1. SUMMARY OF CHANGES

Protocol Amendment 1.4 (dated 05 February 2025) was amended to Amendment 1.5, dated 19 May 2025.

The justification for this amendment is to address regulatory agency feedback and align operations, assessments, and analyses across study sites.

Changes to the protocol included the following:

- Updated medical monitor to reflect current study administrative structure (Table 1)
- Expanded study population to patients with HAE type 3 per regulatory request (Section 4.1, Section 4.3, Section 4.4,Section 6.5, Section 7.1 [Inclusion Criteria #2 and #3], and Appendix 2 [new section])
- Clarified terminology for Treatment Satisfaction Questionnaire for Medication (TSQM-9) (Section 5 and Section 6.1.1)
- Clarified timing of weekly contacts during the Run-in Period (Section 6.1.2.1 and Schedule of Events [Table 4, footnote ‘g’)
- Clarified details regarding attainment of written informed consent from adolescent participants (Section 6.1, Schedule of Events [Table 4, footnotes ‘f’ and ‘g’), Section 7.1 [Inclusion Criteria #1 and #7], Section 12.1, and Section 16.2)
- Clarified sequence of events and timeline for data entry for eligibility review (Section 6.1, Schedule of Events [Table 4, footnotes ‘a’ and ‘b’], and Section 6.1.2.1)
- Modified description of sample types, to remove ‘serum’ or ‘plasma’ where not applicable to define (Schedule of Events [Table 4, footnote ‘i’] and Section 12.8.2)
- Transitioned height measurement at the EOT Visit from the physical examination assessment to the height/weight assessment for adolescent participants (Schedule of Events [Table 4, footnotes ‘j’ and ‘k’], Section 6.1.2.5, Section 12.4, and Section 12.5)
- Added ‘head’ to the list of components included in a physical examination (Schedule of Events [Table 4, footnote ‘k’] and Section 12.5)
- Clarified details of urinalysis assessment to align with study practice (Schedule of Events [Table 4, footnote ‘n’)
- Clarified procedure for clinical laboratory assessment, including safety and pregnancy testing, to allow for more frequent pregnancy testing as per local requirements and also allow for laboratory tests to be initiated by Investigators or designee, assessed locally in parallel with the central laboratory (Schedule of Events [Table 4], footnotes ‘n’ and ‘o’], Section 12.8.1, and Section 15)
- Added a tabular presentation of PK sample collection time points to improve clarity (Schedule of Events [Table 4, removed footnote ‘o’ and added table cross-reference in row titled, ‘PK sampling’] and Table 5 [new table])

- Increased number of adults and adolescent participants planned to undergo the full sampling schedule for PK assessments to support a robust PK evaluation of the study drug and clarified participants planned to undergo the less frequent PK sampling schedule (Table 5 [new table] and Section 12.8.2)
- Clarified timing of PK sample collection for participants entering Study PHA022121-C307 (Table 5 [new table] footnote 'a' and Section 12.8.2)
- Clarified procedures for ePRO retraining and repeat assessments (Schedule of Events [Table 4, footnotes 's' and 't'] and Section 6.1.2.2)
- Added a minimum body weight eligibility requirement for adolescent participants to support consistent study drug exposure across the study population. (Section 7.1, [Inclusion Criterion #1])
- Clarified procedure for C1q testing for determining the eligibility of participants with type 1/2 HAE (Section 7.1 [Inclusion Criterion #3a])
- Added C4 assessment to diagnostic eligibility requirements to support a definitive HAE type 1/2 diagnosis (Section 7.1 [Inclusion Criterion #3a])
- Clarified study treatment discontinuation and participant withdrawal (Section 7.3)
- Clarified guidance on use of medications that are substrates of P-gp and BCRP transporters based on recent data from a drug-drug interaction study conducted by the Sponsor (Section 8.2.3)
- Removed specific volume of water for study drug administration because this level of detail can be provided outside of the protocol (Section 9.4)
- Clarified language related to HAE attacks (Section 10.1 and Section 10.2)
- Added option for an Investigator's designee to perform select responsibilities, as applicable (Section 10.2 and Section 12.8.1)
- Clarified the procedure for the ECG assessment and recording (Section 12.7)
- Reinstated language regarding AESIs that had been moved in a previous amendment to improve clarity (Section 12.9.1.3.1)
- Updated terminology for relationship of AEs to study drug from "Unrelated" to "Not related" (Section 12.9.2)
- Added details regarding timing and severity of AEs to improve clarity (Section 12.9.3.1 and Section 12.9.3.2)
- Updated handling and reporting of on-study pregnancies (Section 12.10)
- Clarified reporting study treatment errors, including misuse, abuse and overdose procedures (Section 12.11)
- Redefined ICEs and corresponding data handling per regulatory request (Section 13.5.1 and Section 13.5.2)
- Modified testing hierarchy of a study endpoint per regulatory request (Section 13.11)

In addition, changes made in this amendment included several minor editorial and administrative changes and the protocol synopsis was updated in alignment with changes made to the protocol body.

APPENDIX 2. HAE TYPE 3 (HAE-NL-C1INH) GENE MUTATIONS FROM THE US HAEA AND WAO/EAACI GUIDELINES

HAE-nL-C1INH Sub-type	Gene/Underlying mutation	Supporting Literature
HAE-FXII	F12, the gene encoding coagulation FXII9	Cichon, 2006
HAE-PLG	PLG, the gene encoding plasminogen	Bork, 2018
HAE-ANGPT1	ANGPT1, the gene encoding angiopoietin-1	Bafunno, 2018
KNG1	kininogen 1 gene	Bork, 2019
HAE-MYOF	Myoferlin gene	Ariano, 2020
HAE-HS3ST6	Heparan sulfate 3-O-sulfotransferase 6 gene	Bork, 2021

Sources: (Busse, 2021, Maurer, 2022)